

(12)
United States Patent
Hojjat et al.

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(45) **Date of Patent:** **Aug. 26, 2025**

(54) **ANTENNA ELEMENT WITH SPATIAL FILTERING PROPERTY USING FREQUENCY SELECTIVE UNIT CELL BUILDING BLOCKS**

(58) **Field of Classification Search**
CPC H01Q 15/0013; H01Q 1/246; H01Q 21/26; H01Q 15/008; H01Q 15/006
See application file for complete search history.

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(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 148 days.

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(22) Filed: **Mar. 22, 2023**

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Related U.S. Application Data

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(51) **Int. Cl.**
H01Q 15/00 (2006.01)
H01Q 1/24 (2006.01)
H01Q 1/48 (2006.01)
H01Q 9/16 (2006.01)

(52) **U.S. Cl.**
CPC **H01Q 15/0013** (2013.01); **H01Q 1/246** (2013.01); **H01Q 1/48** (2013.01); **H01Q 9/16** (2013.01)

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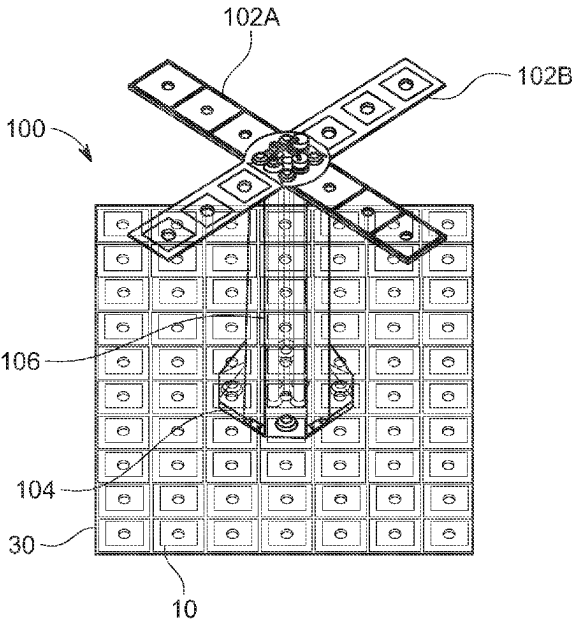
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(57) **ABSTRACT**
An antenna element for use in a cellular base station antenna has a feed network and at least two radiating arms at the top of the network and arranged at an approximately 90 degree position relative to one another. The at least two radiating arms are constructed from a plurality of substantially identically dimensioned unit cells.

15 Claims, 34 Drawing Sheets



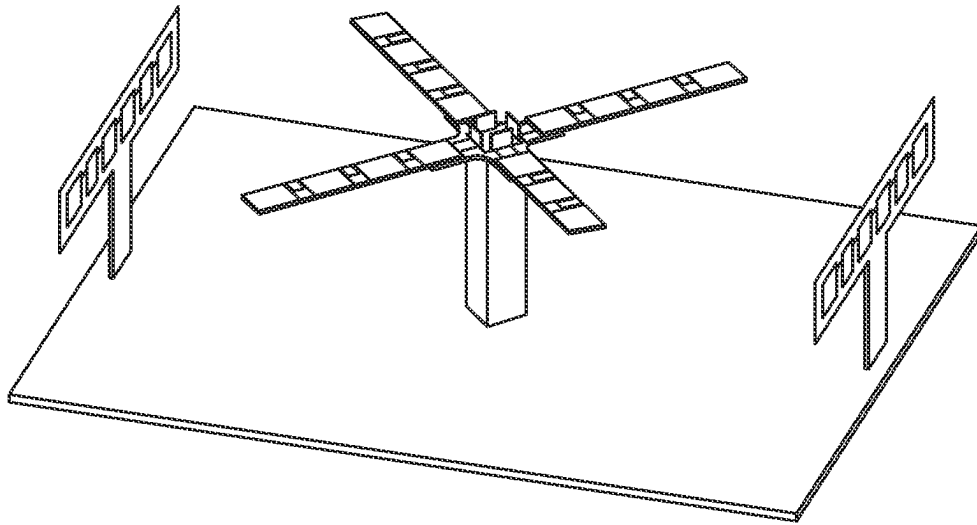


FIG. 1
(PRIOR ART)

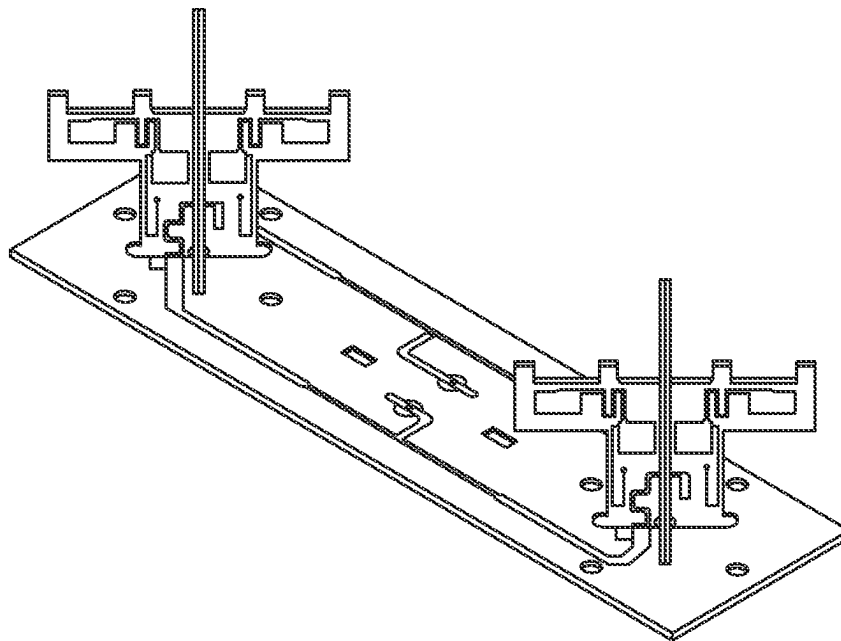


FIG. 2
(PRIOR ART)

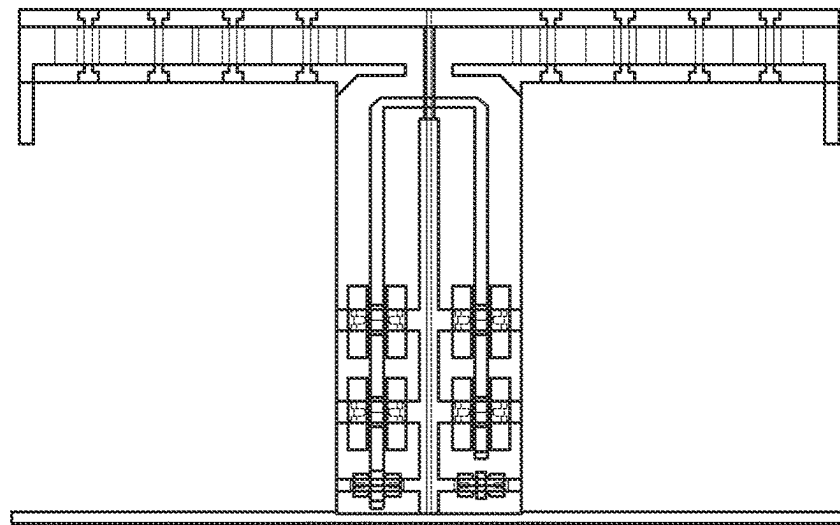


FIG. 3
(PRIOR ART)

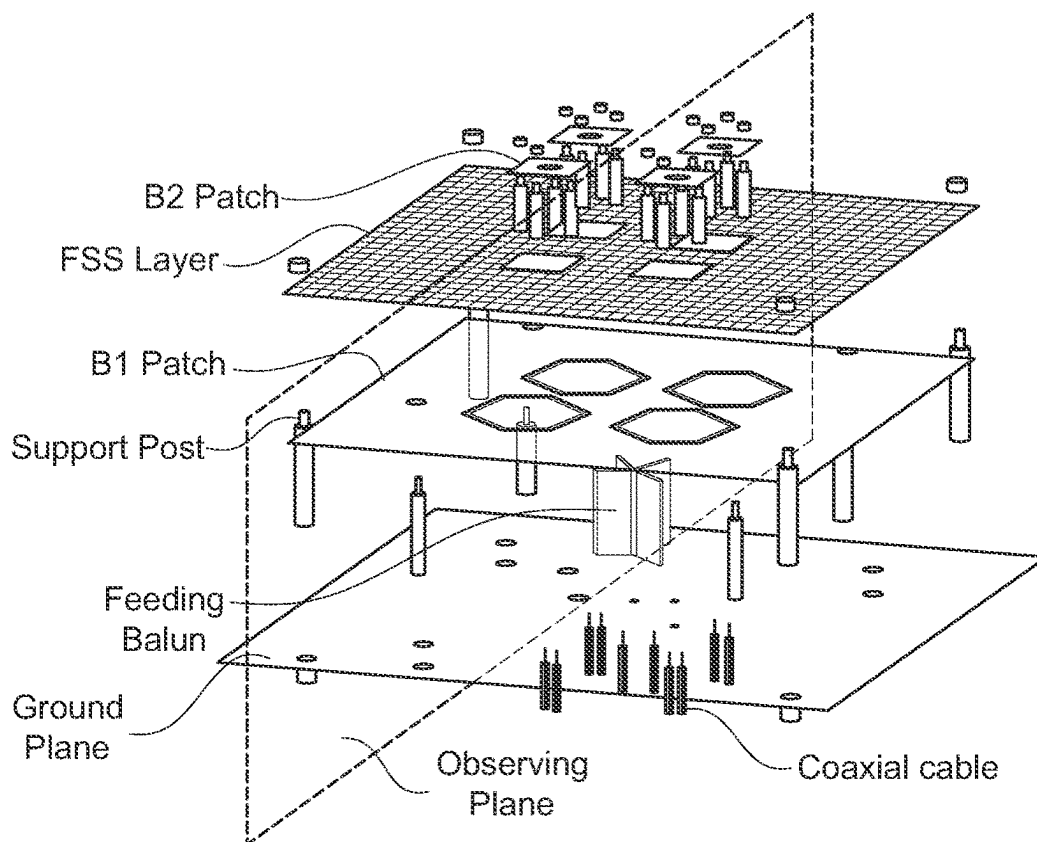


FIG. 4
(PRIOR ART)

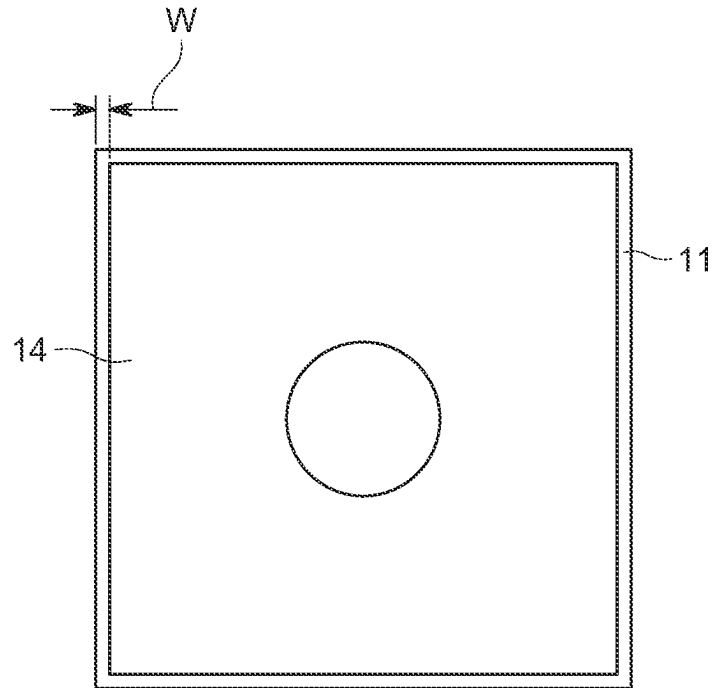


FIG. 5A1

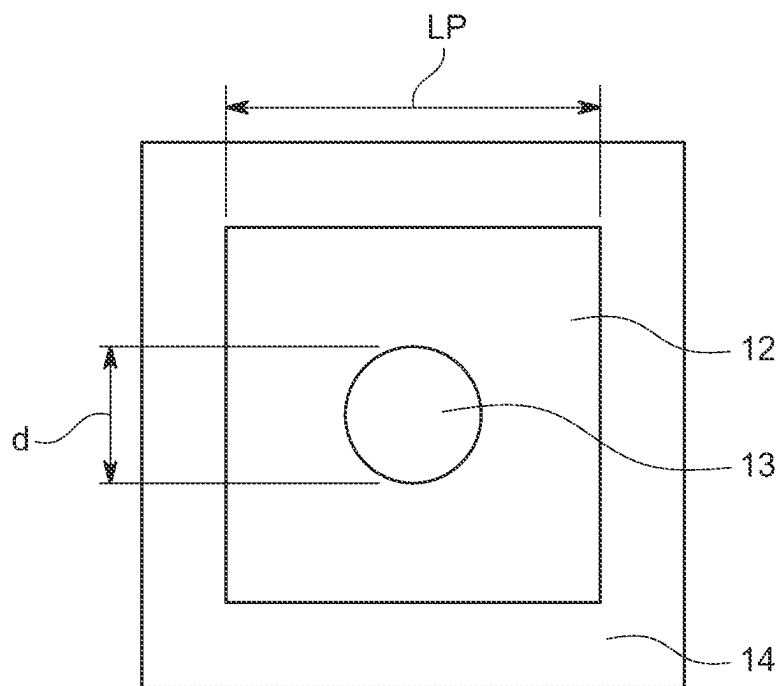
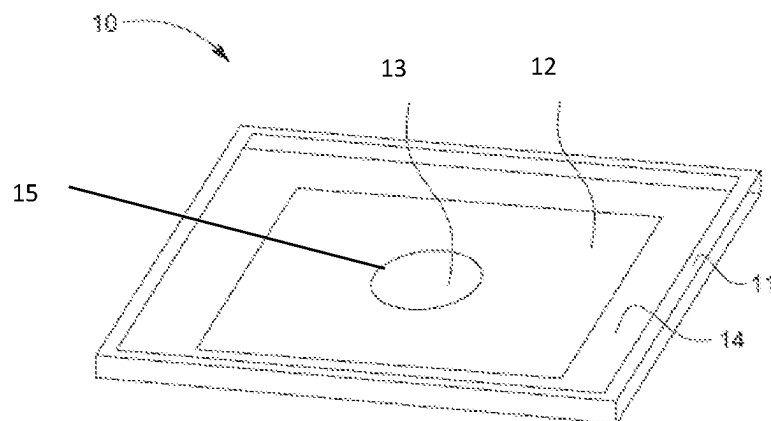
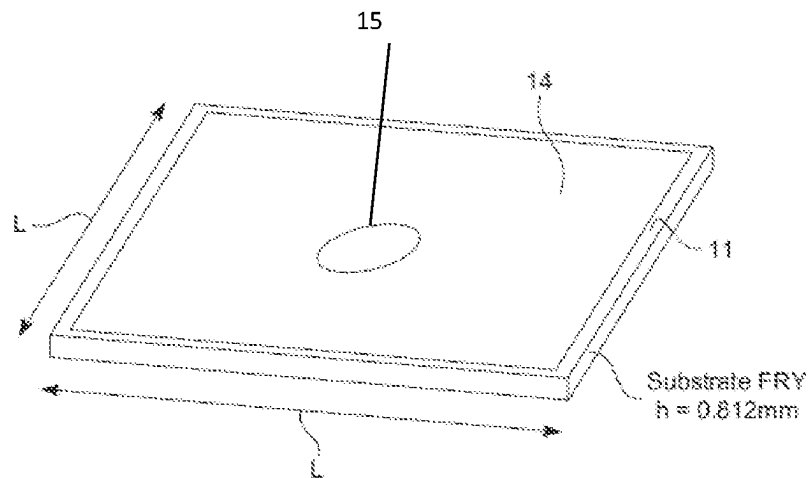


FIG. 5A2



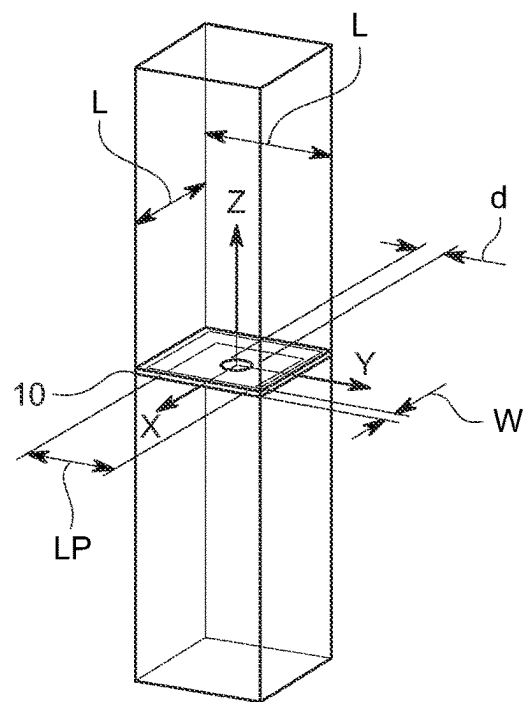


FIG. 5B

Name	X [GHz]	Y
m1	3.2980	-14.6641
m2	3.7980	-15.1903

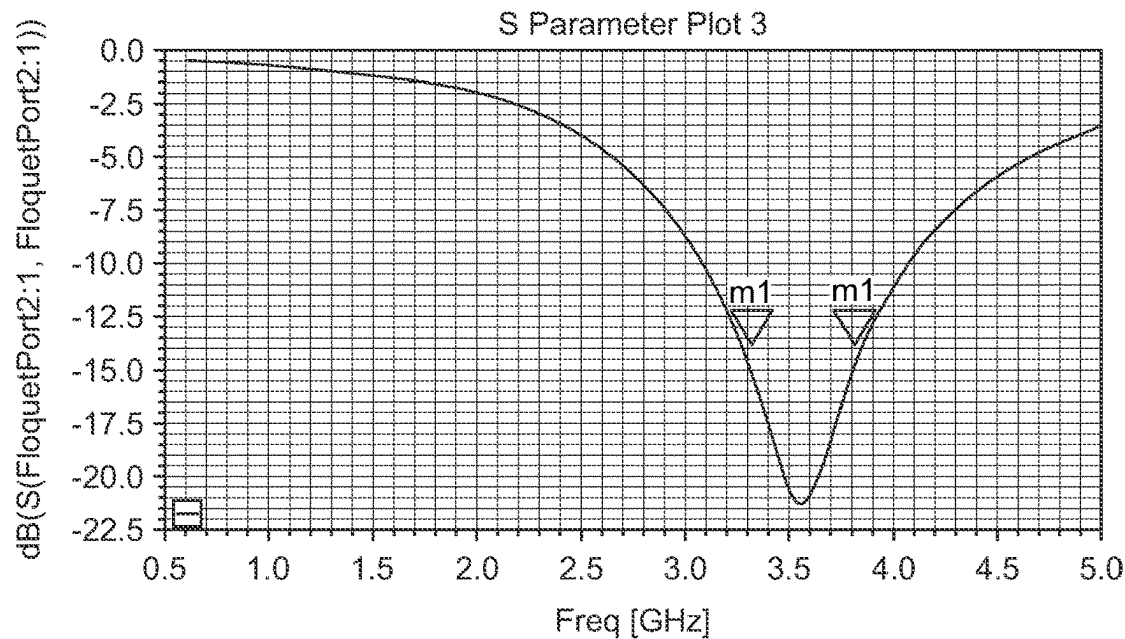


FIG. 5C

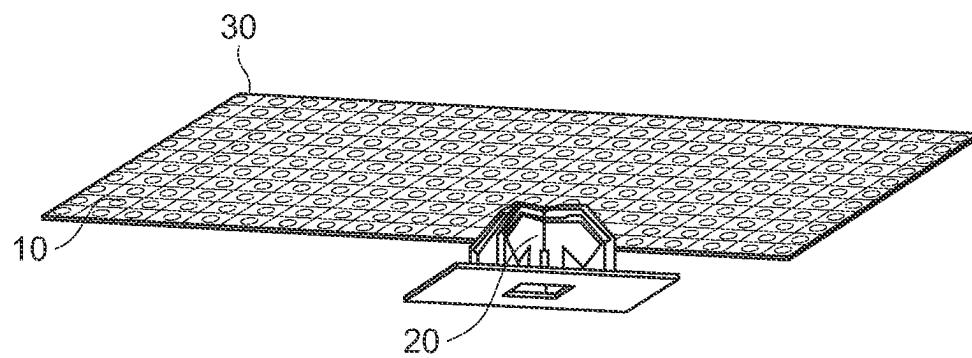


FIG. 6A

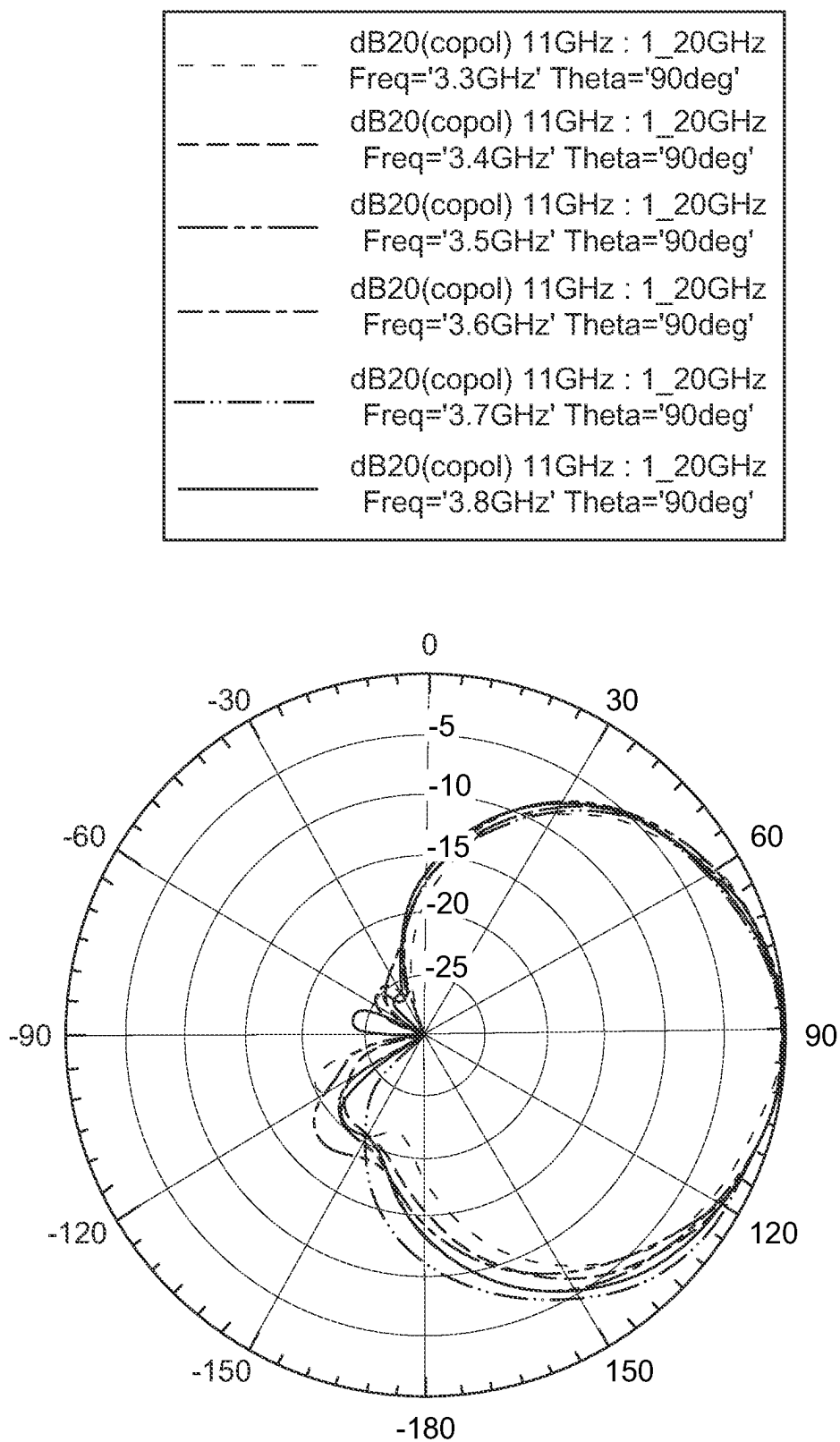


FIG. 6B

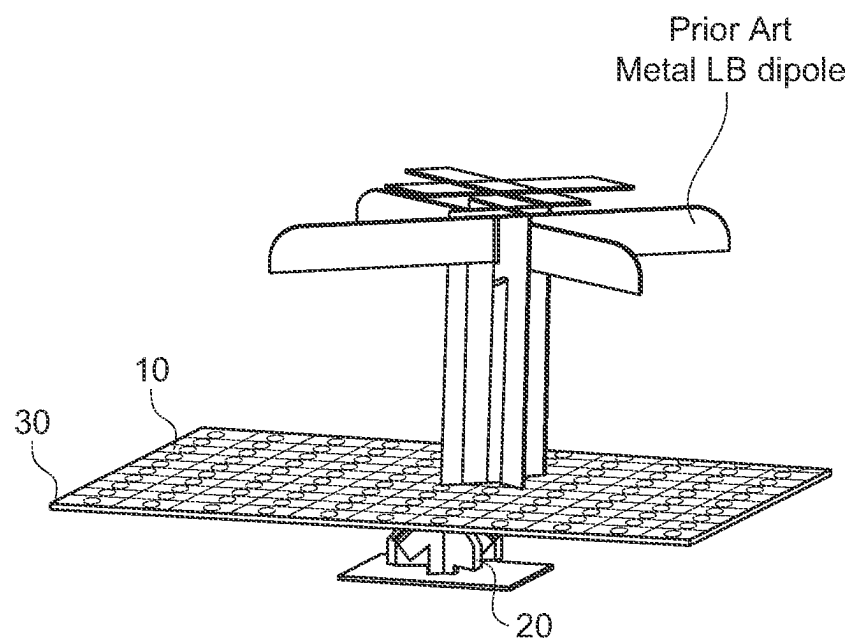


FIG. 7A
(PRIOR ART)

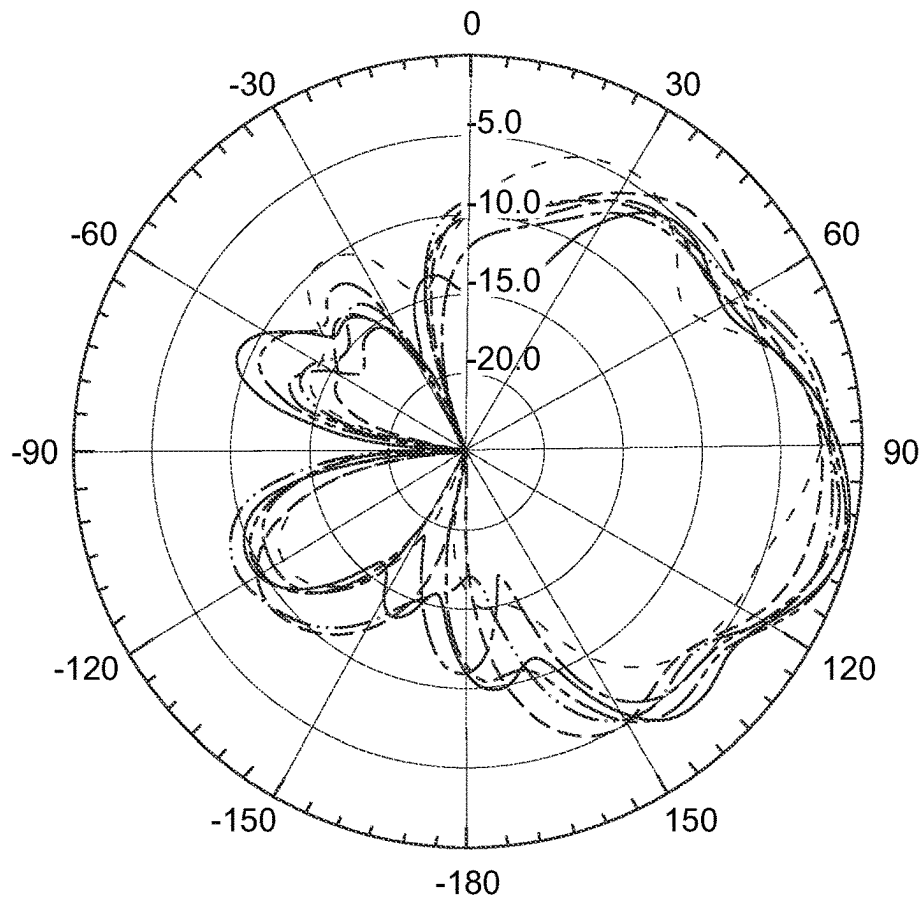
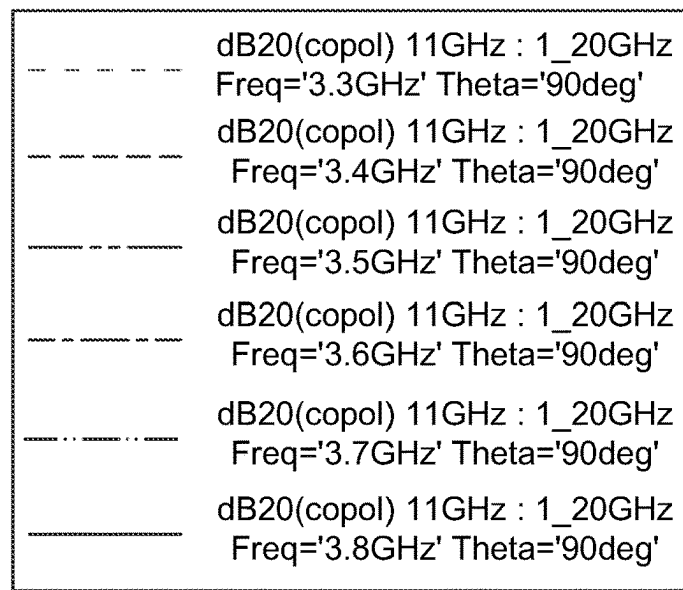


FIG. 7B
(PRIOR ART)

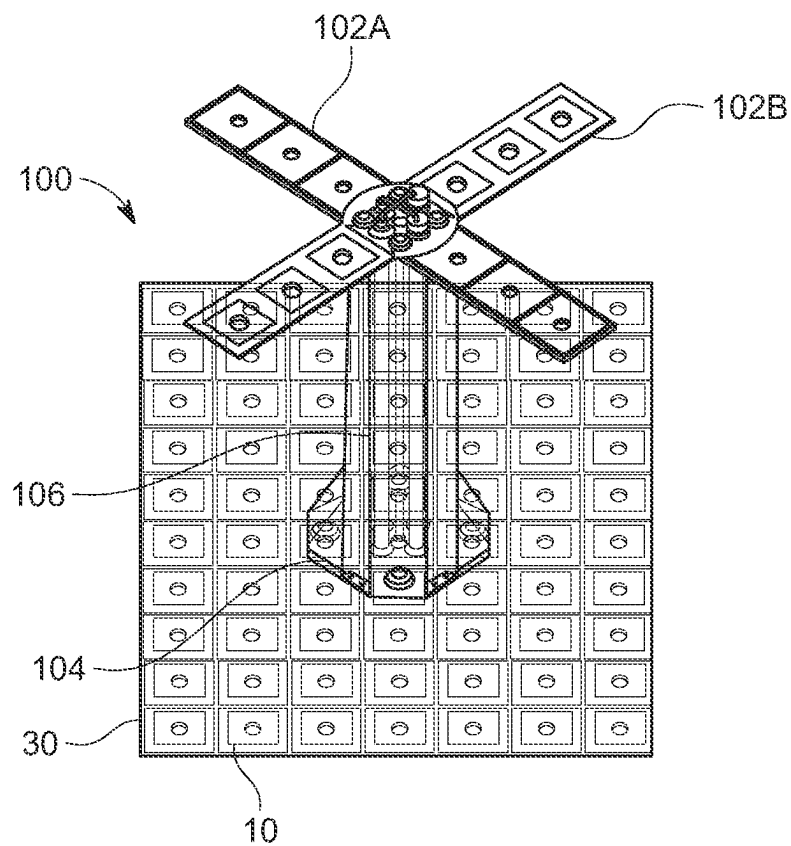


FIG. 8A

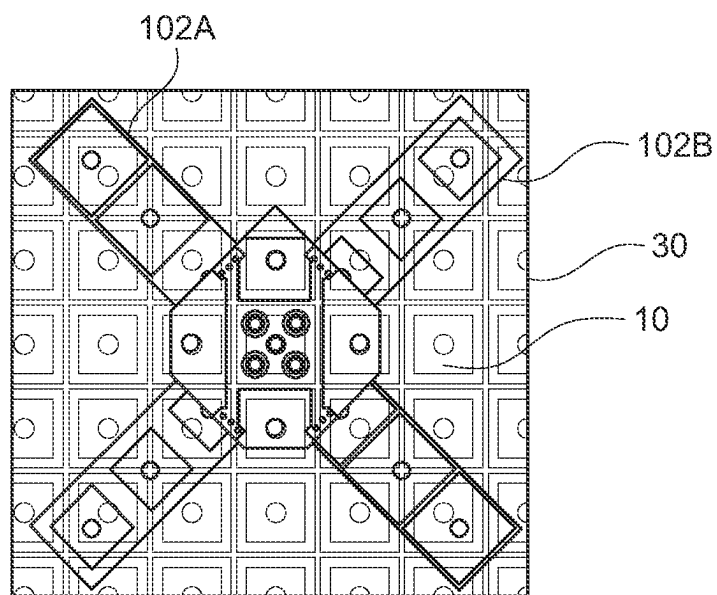


FIG. 8B

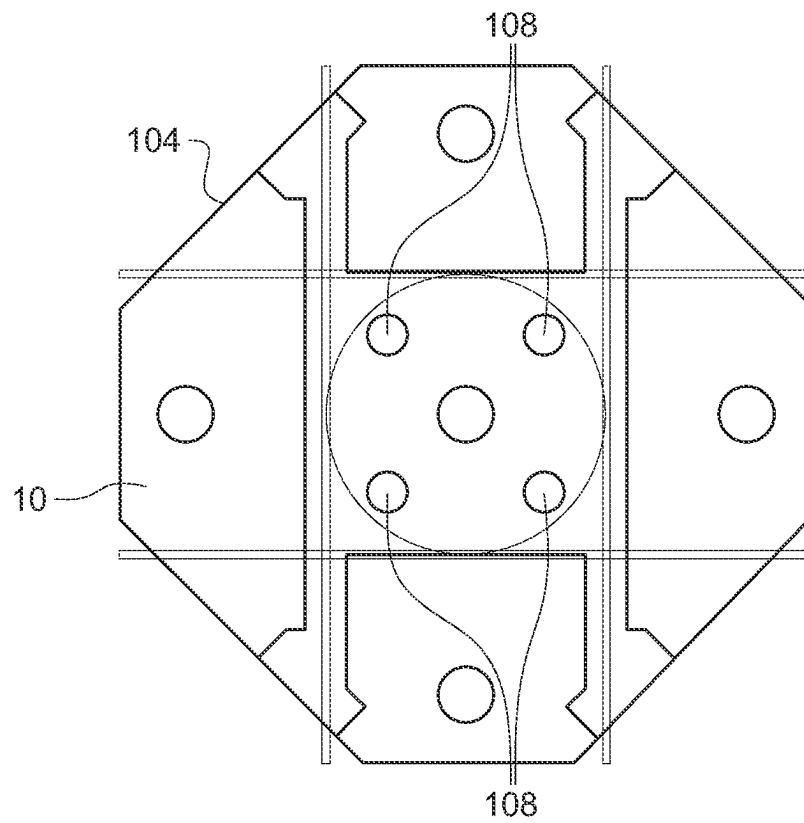


FIG. 8C

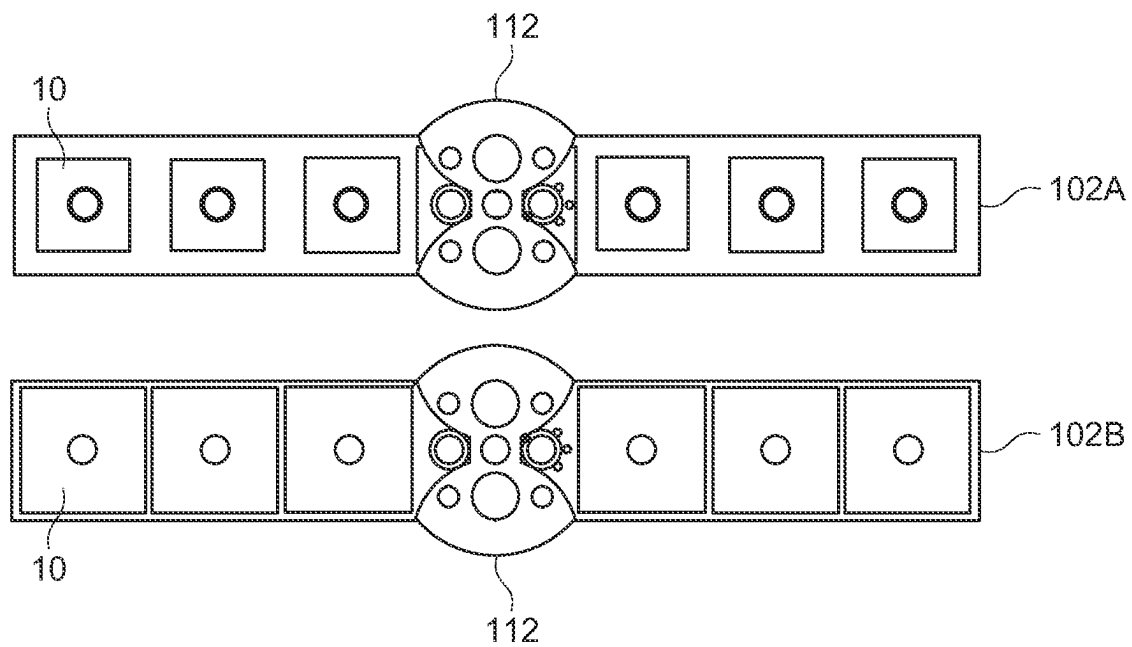


FIG. 8D

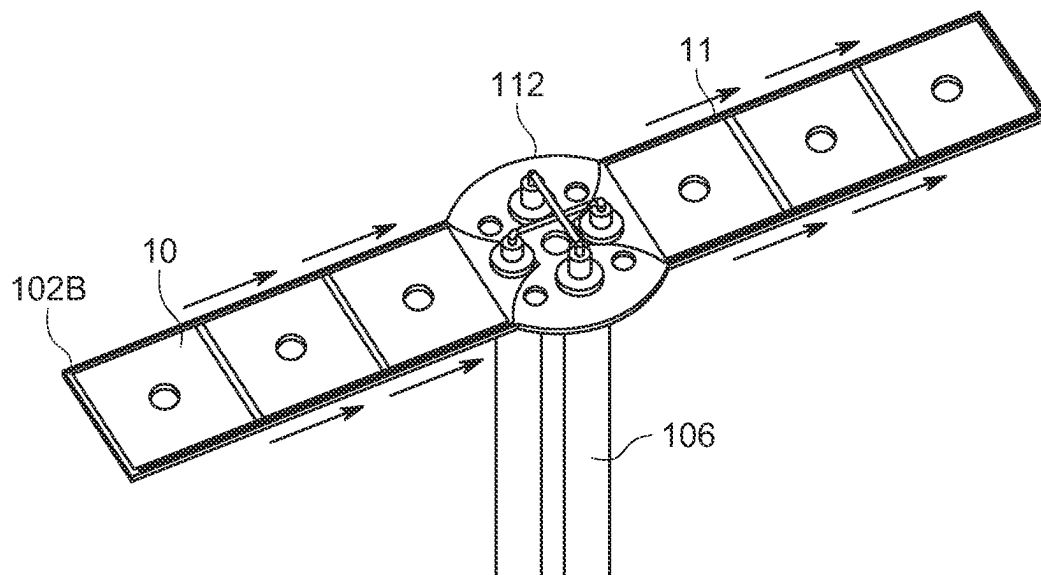


FIG. 8E

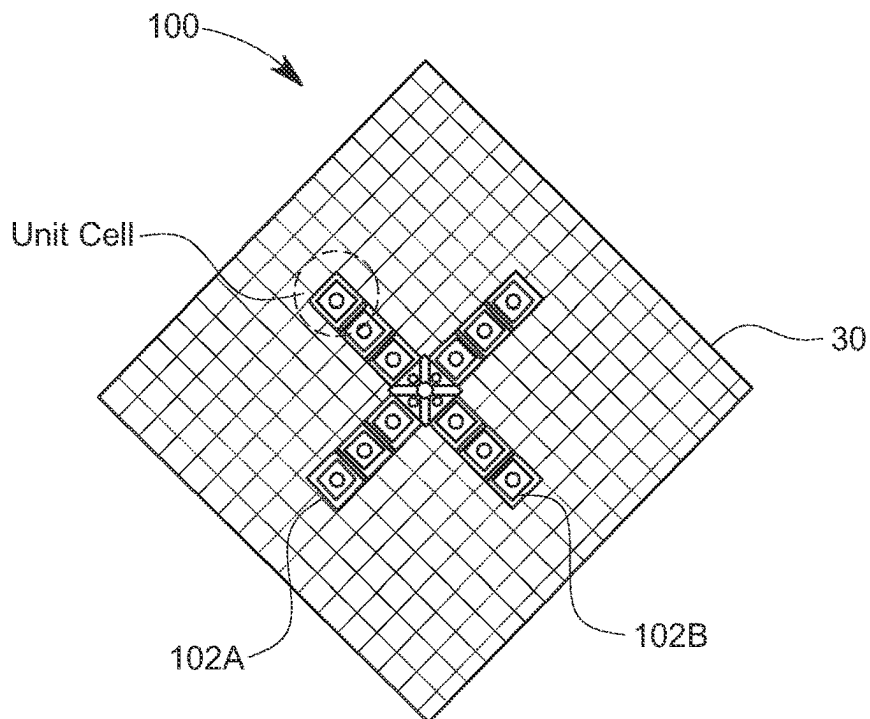


FIG. 9A

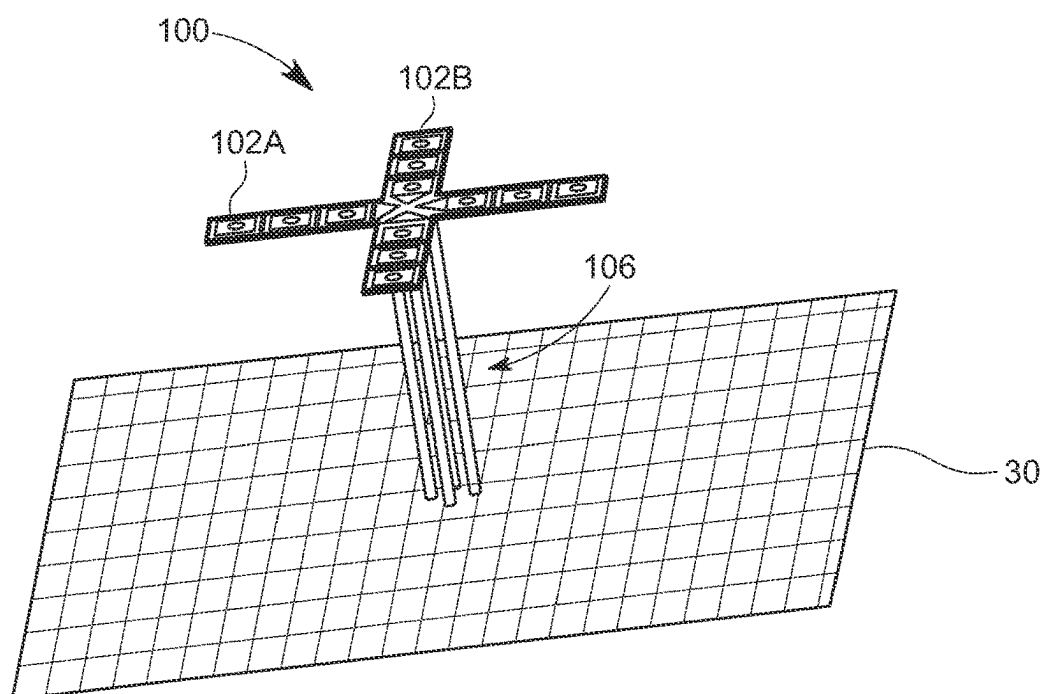


FIG. 9B

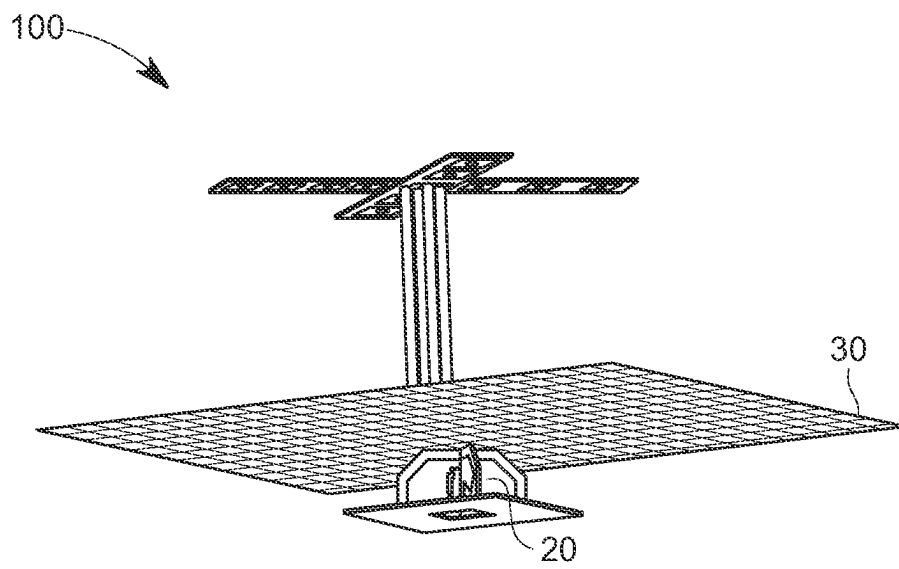


FIG. 10A

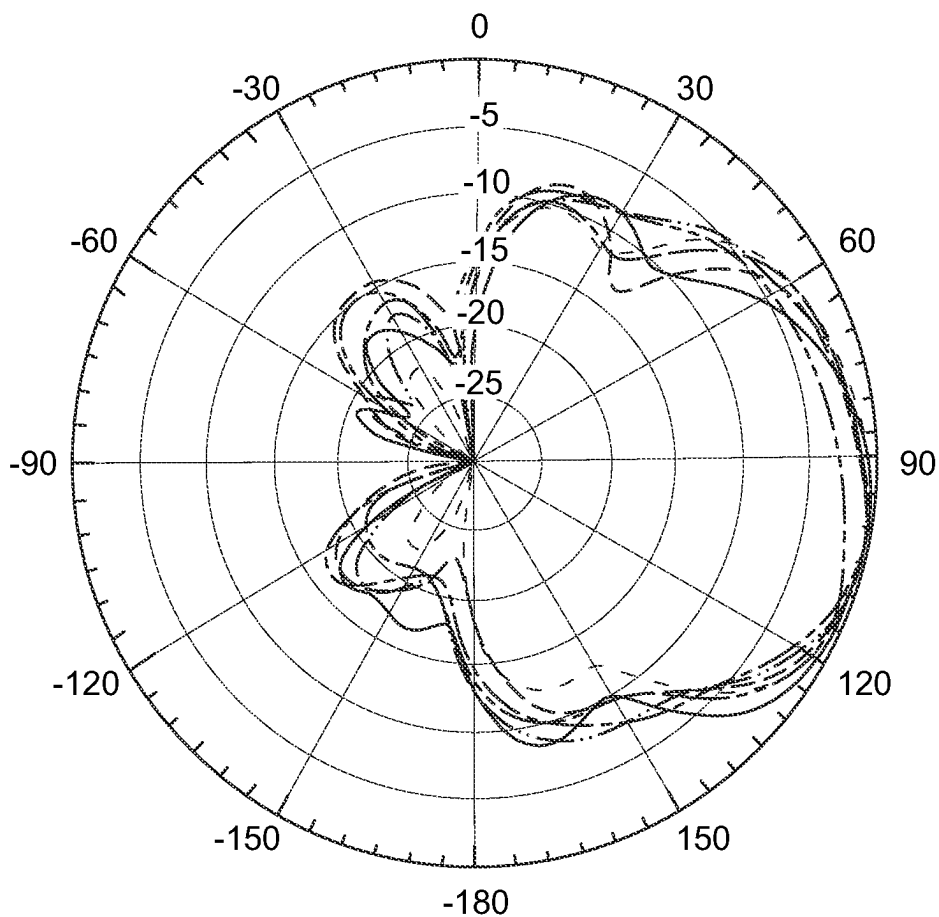
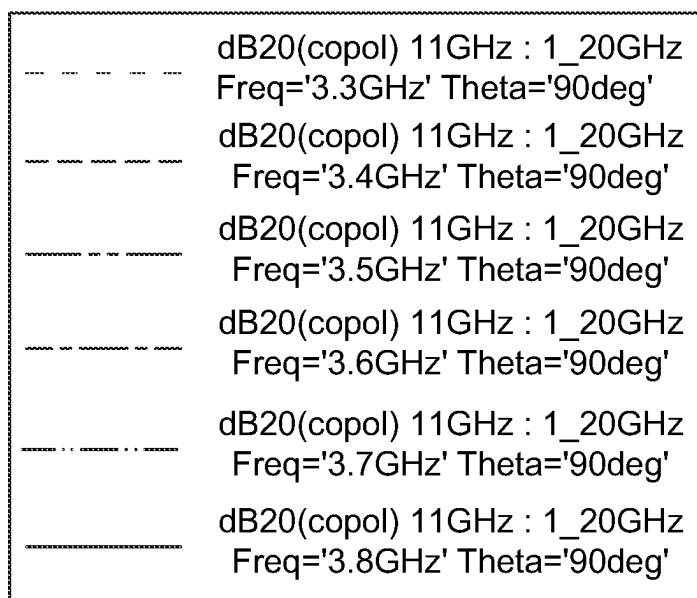


FIG. 10B

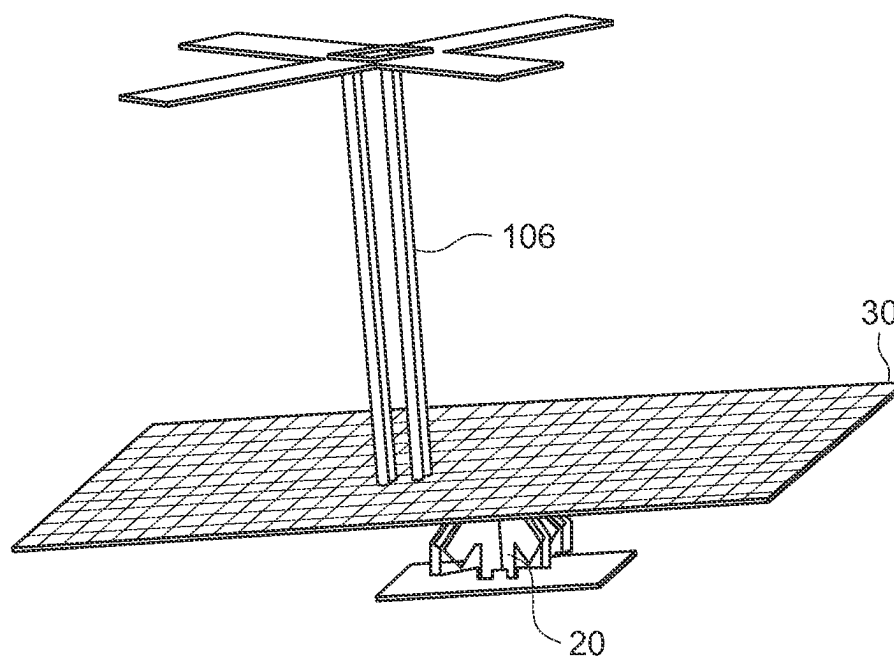


FIG. 10C

Prior Art Metal ARMS Used on NOVEL
Feed Network For LB dipole

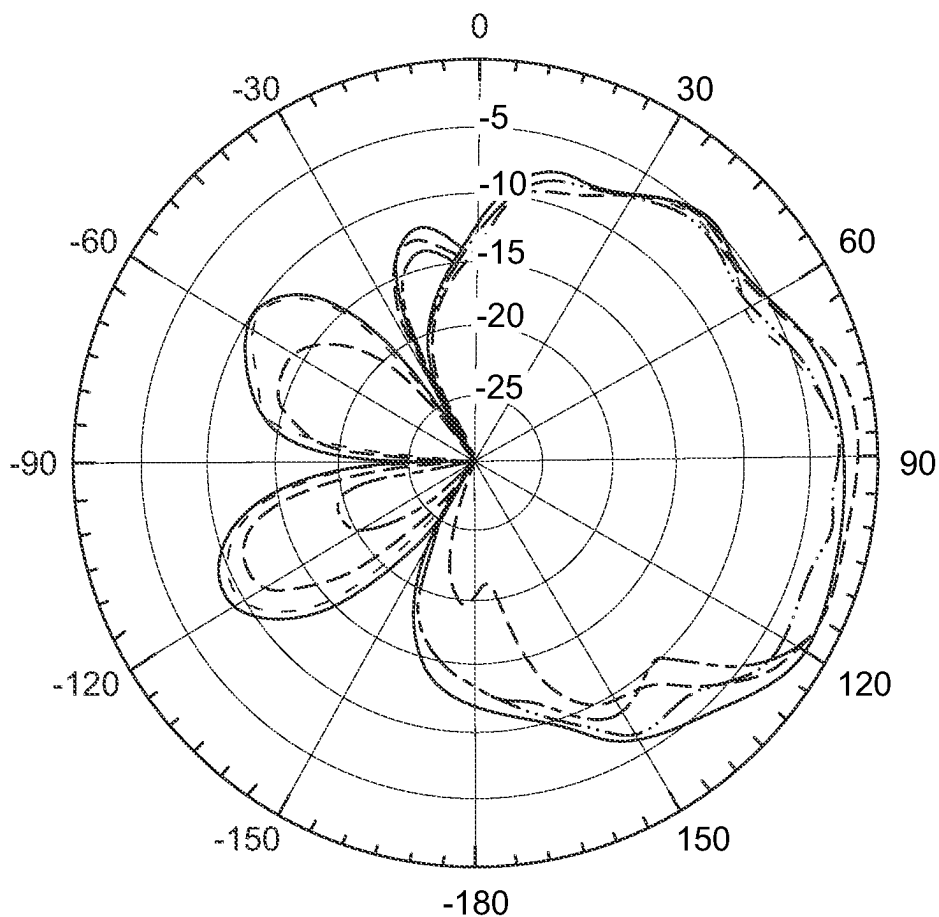
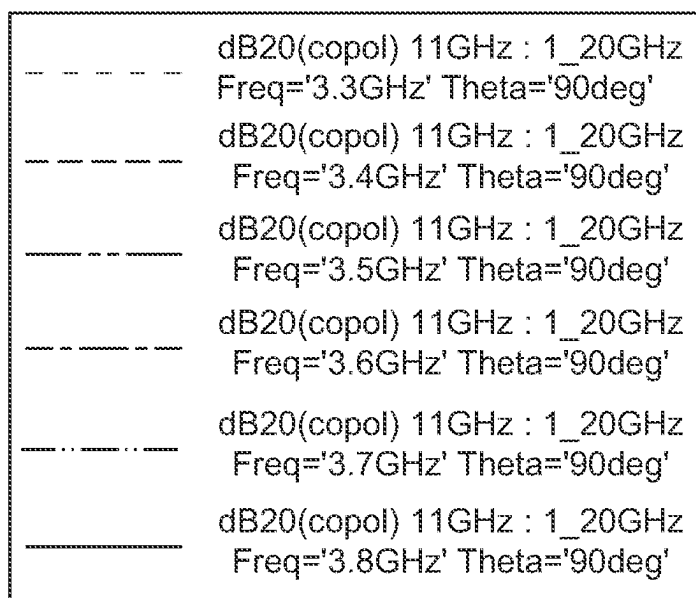


FIG. 10D

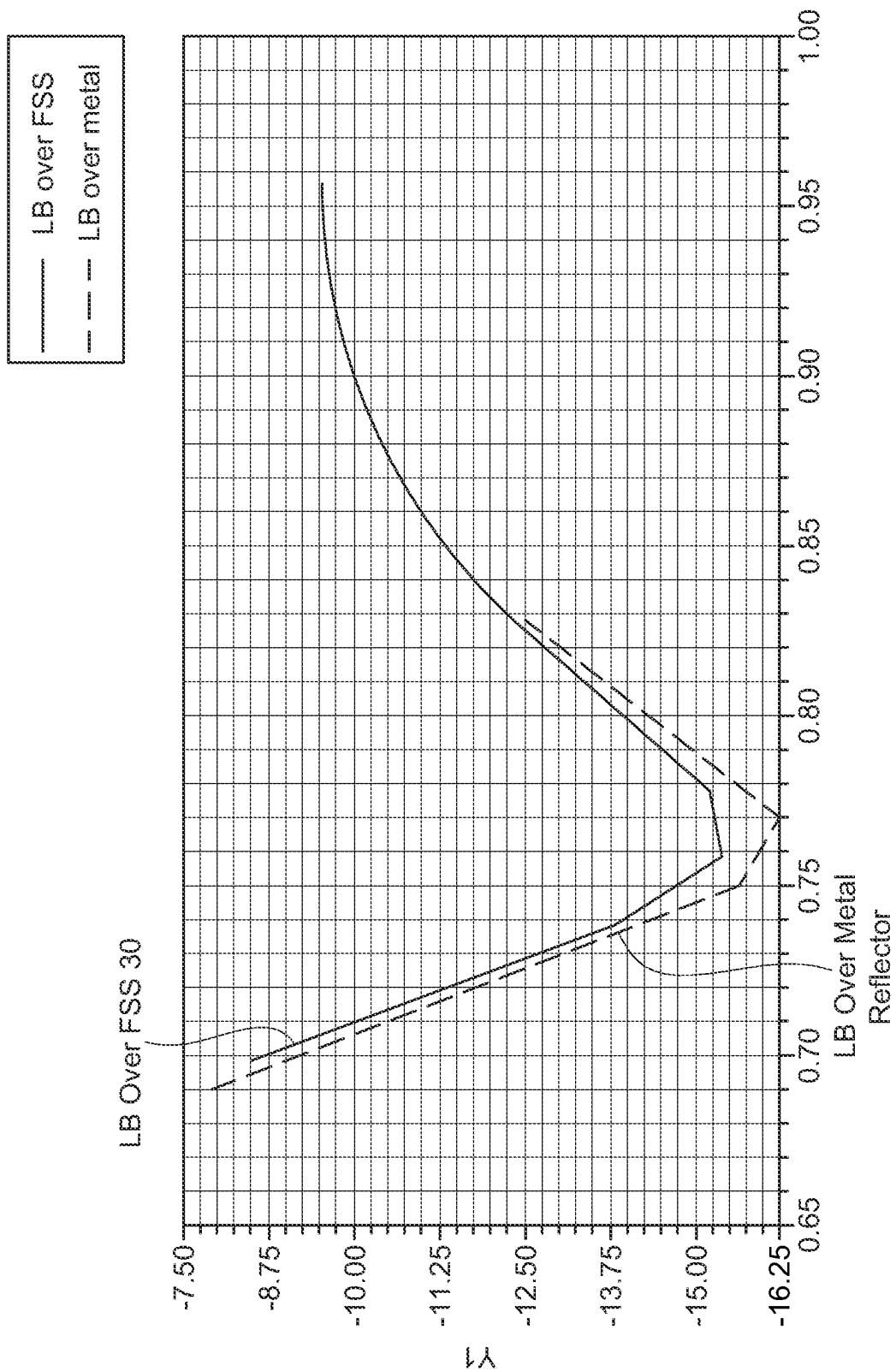


FIG. 11

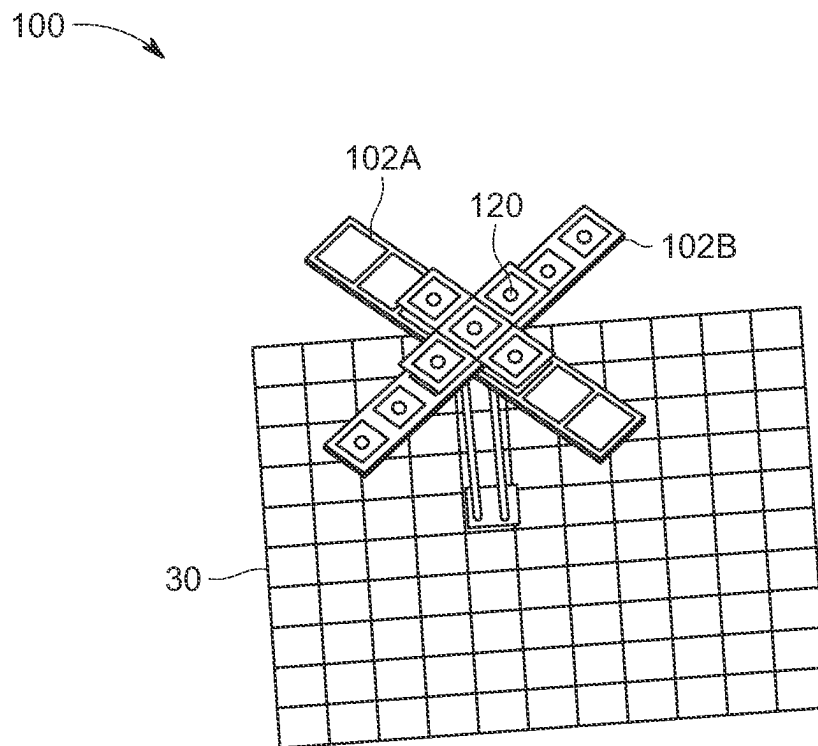


FIG. 12A

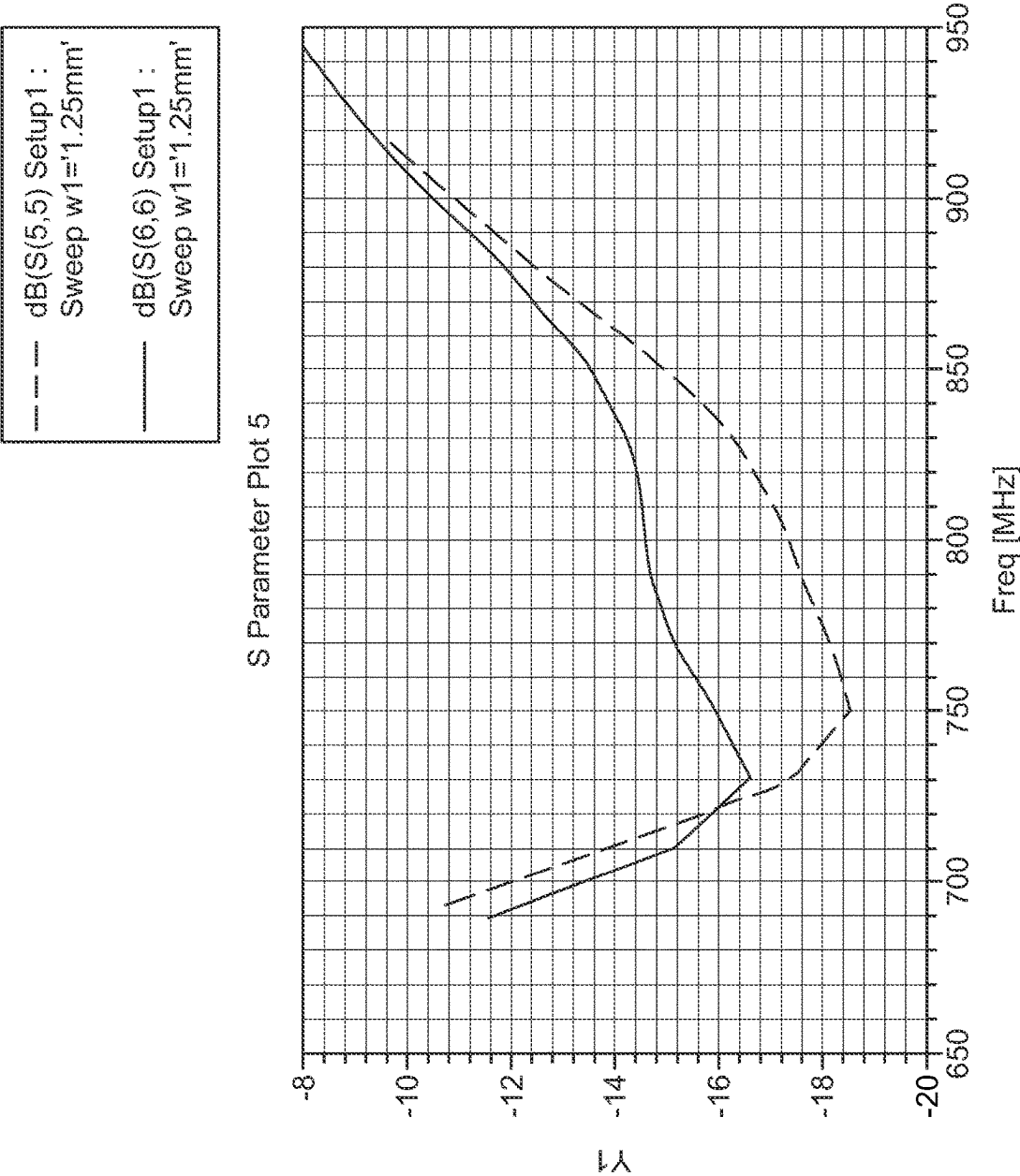


FIG. 12B

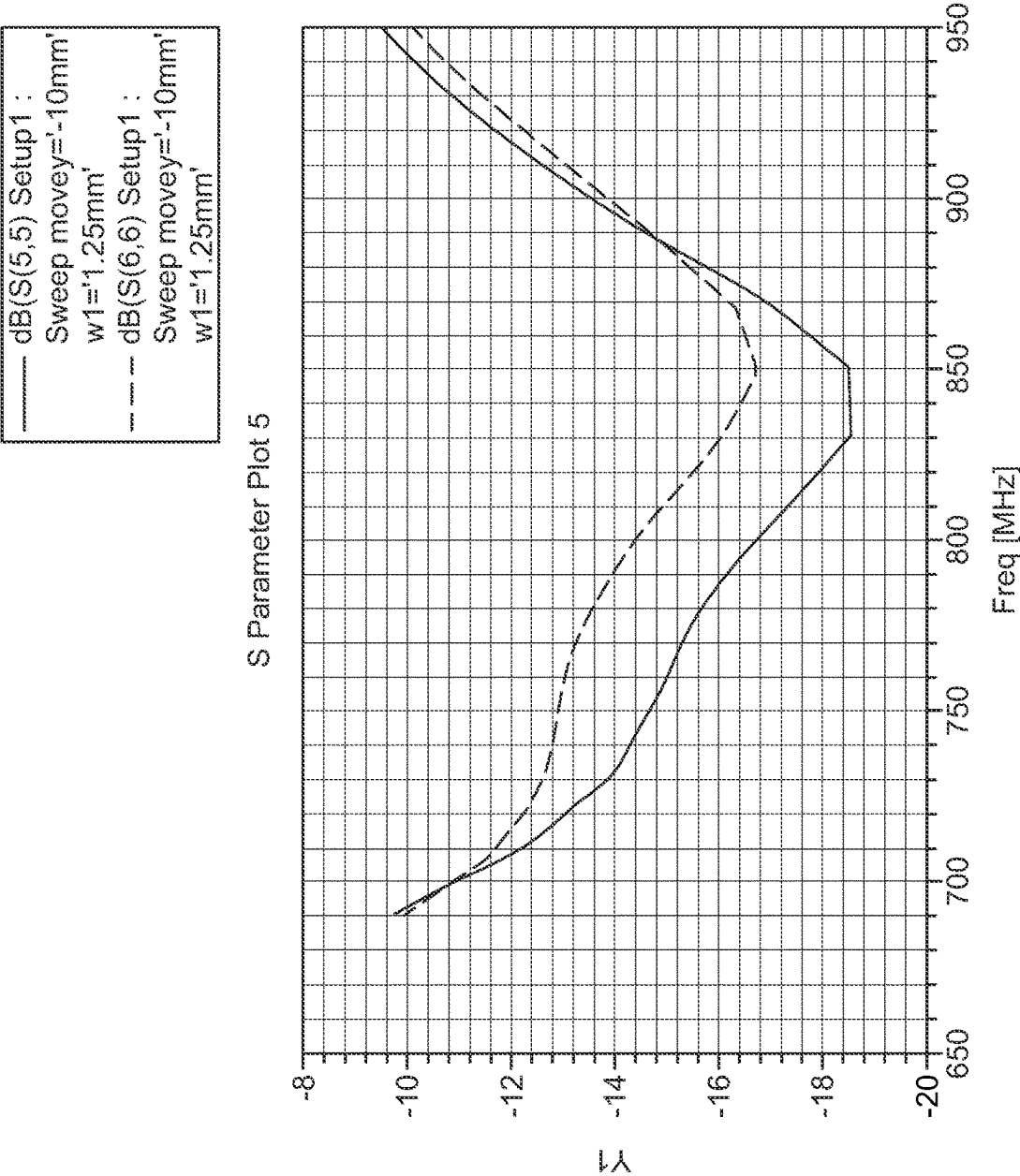


FIG. 12C

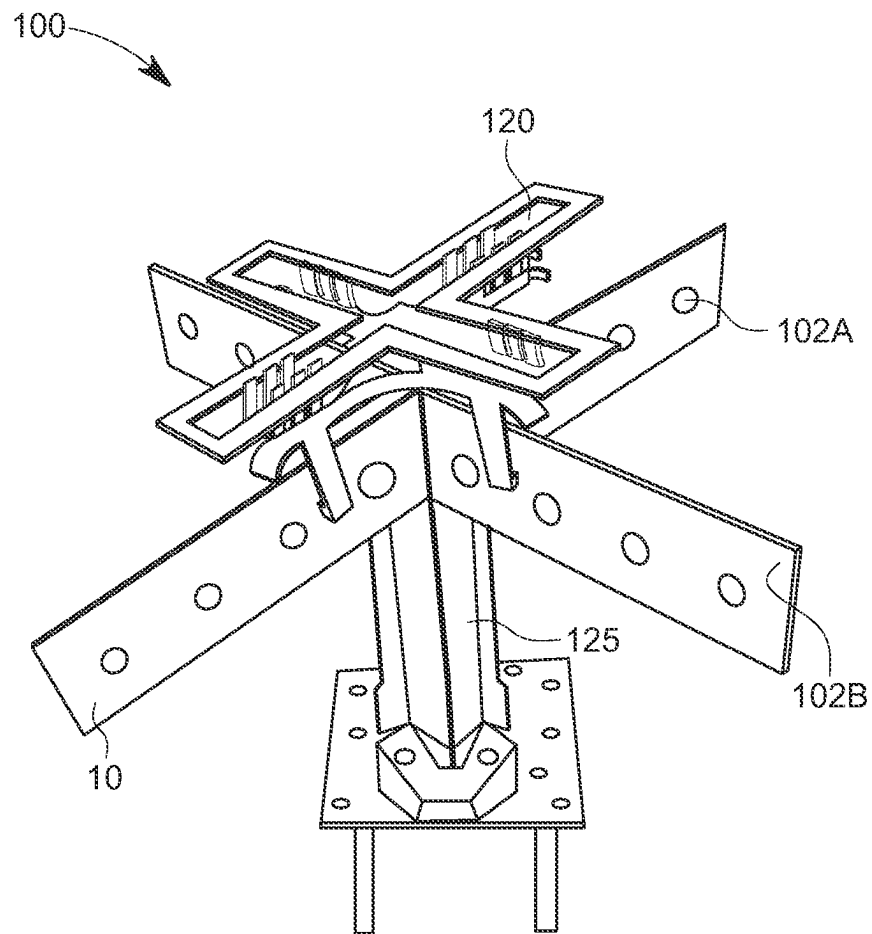


FIG. 13

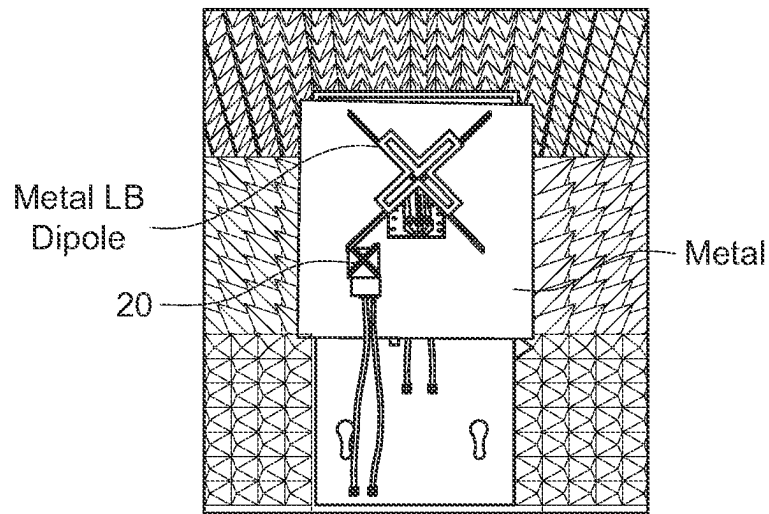


FIG. 14A
(PRIOR ART)

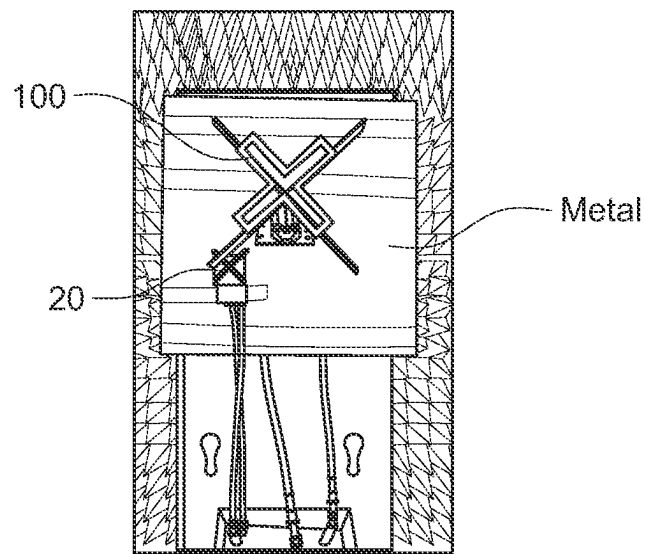


FIG. 14B

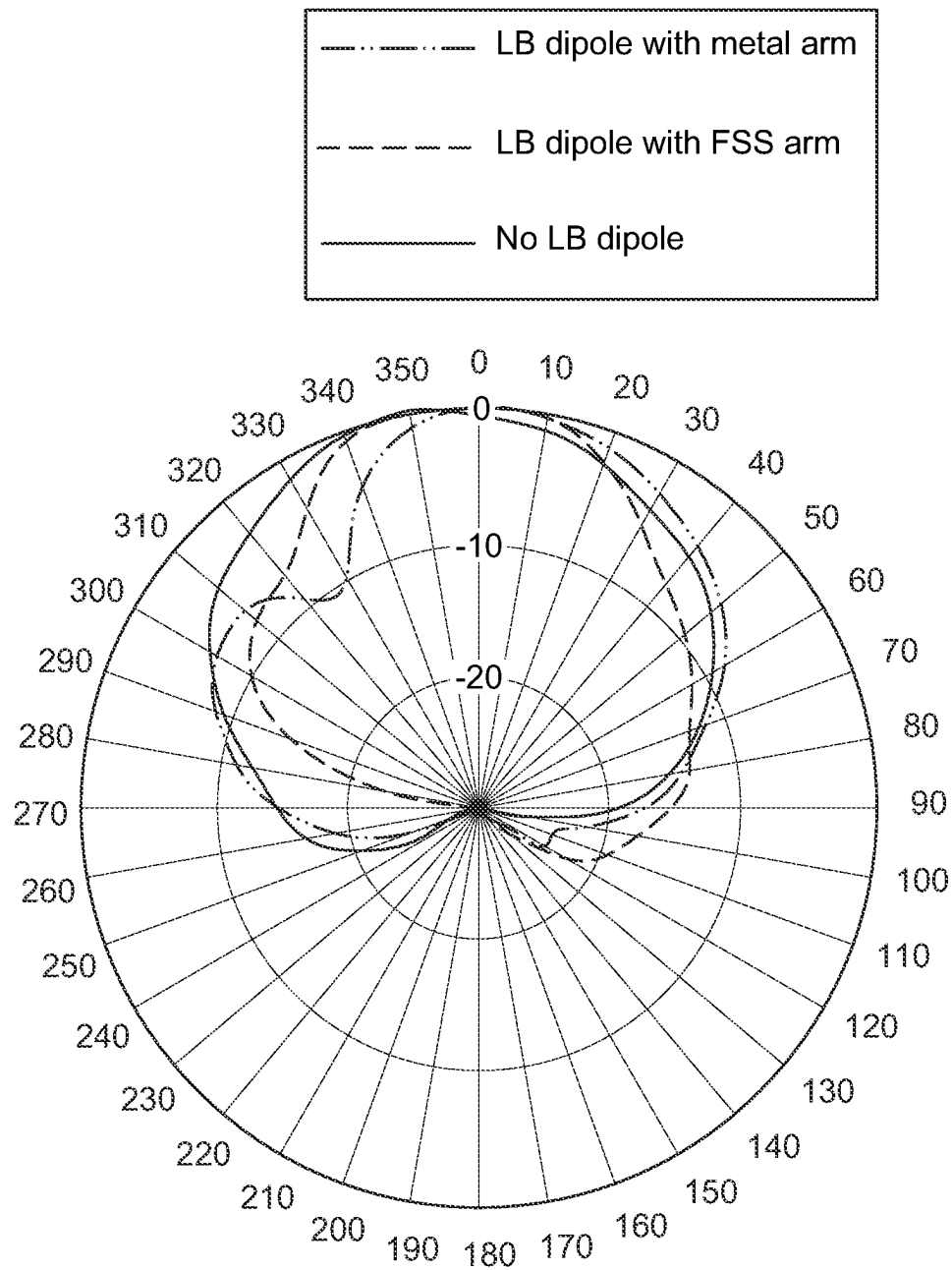


FIG. 15
(PRIOR ART)

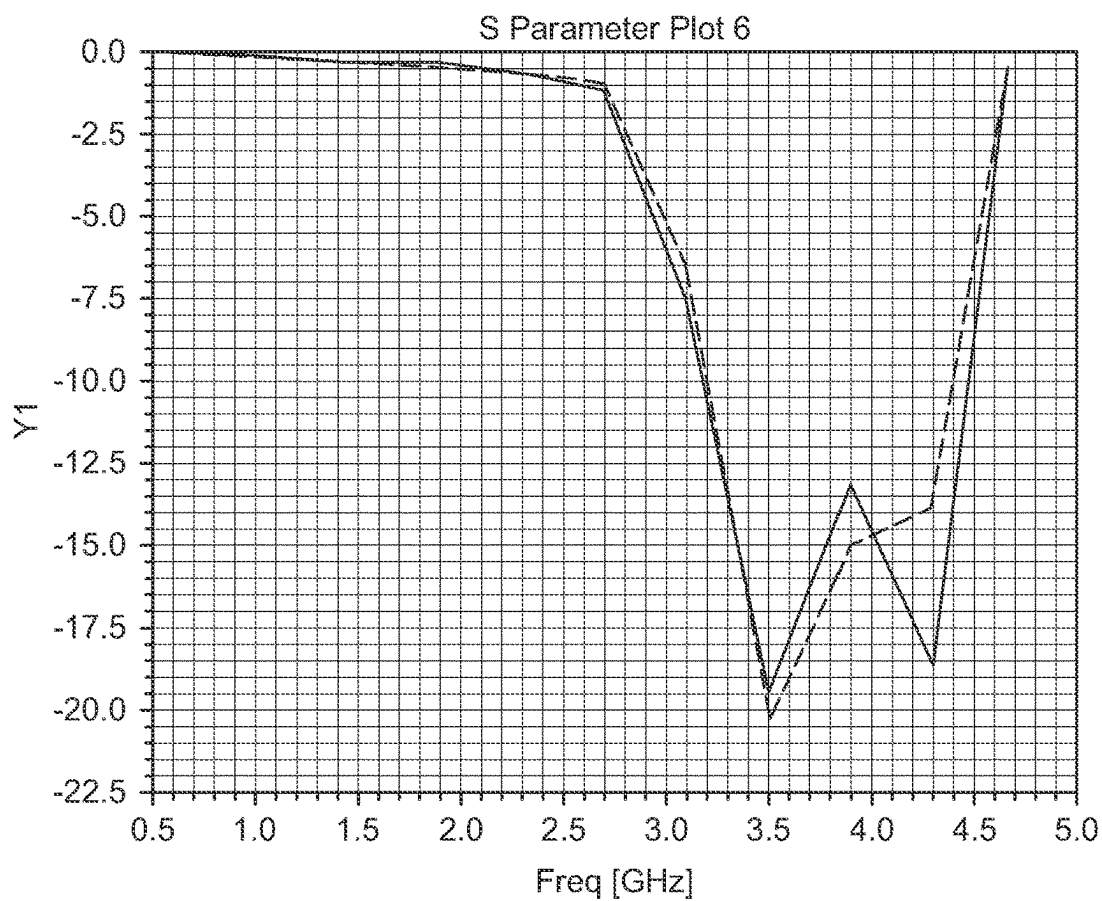
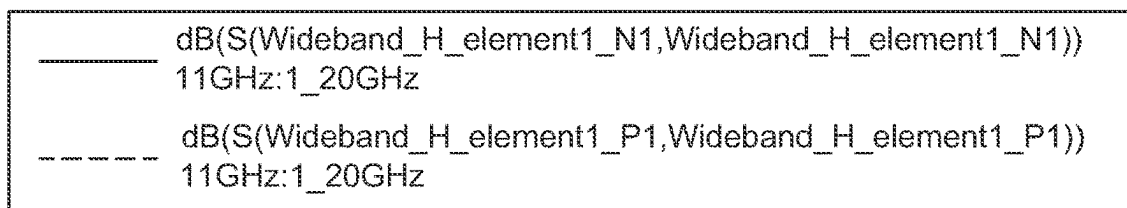
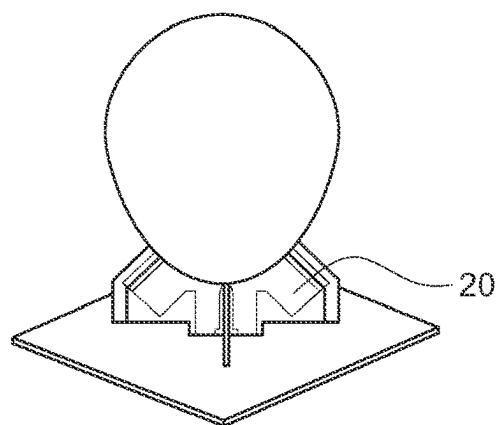


FIG. 16A

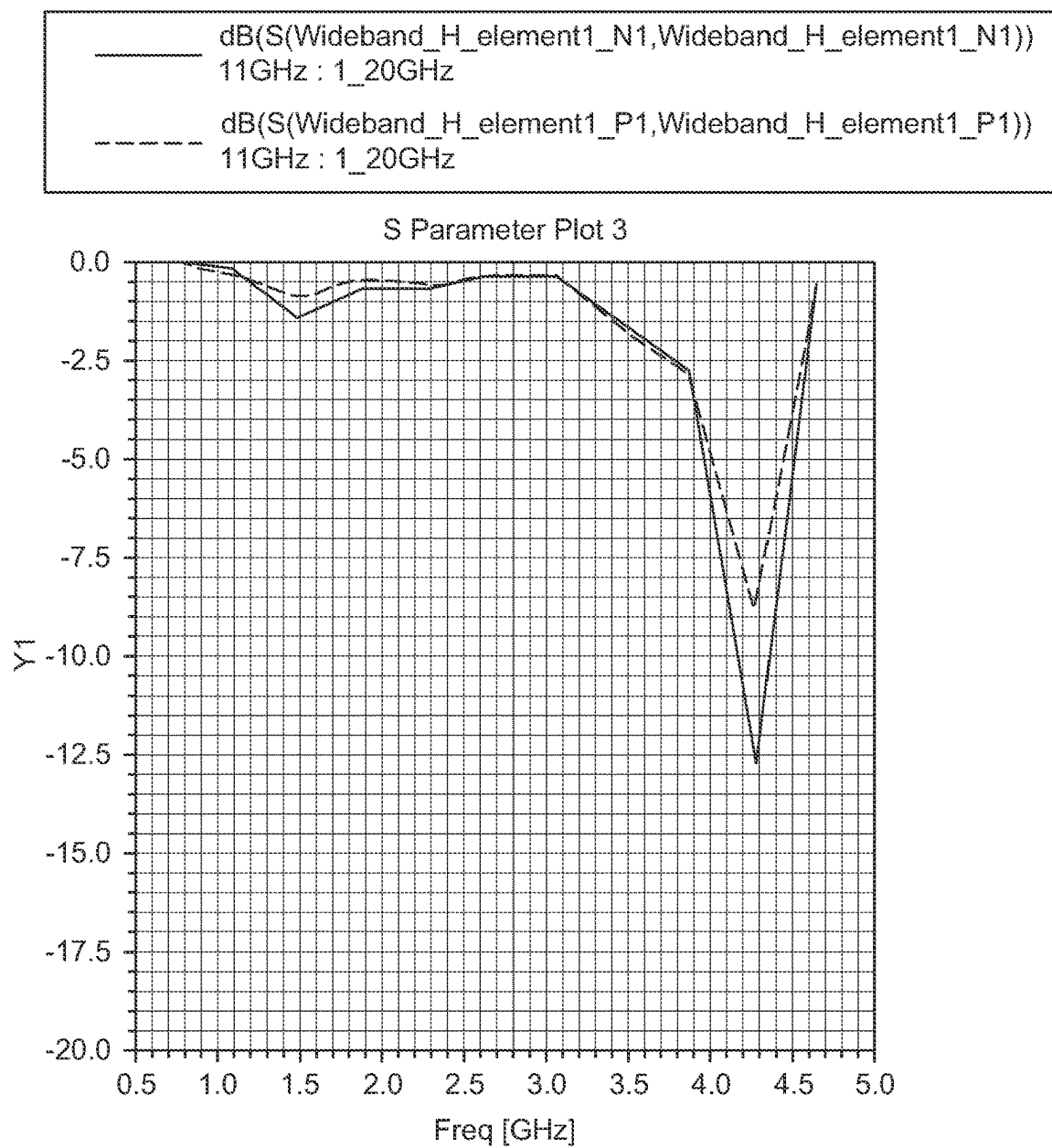
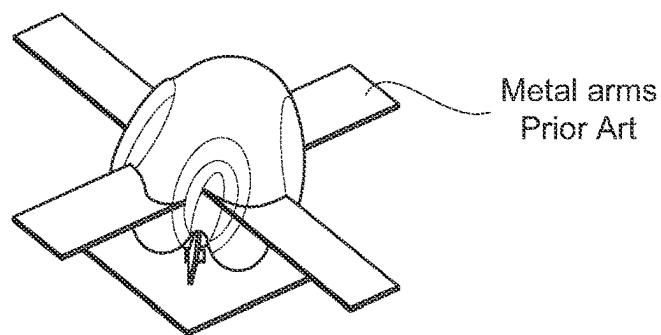


FIG. 16B
(PRIOR ART)

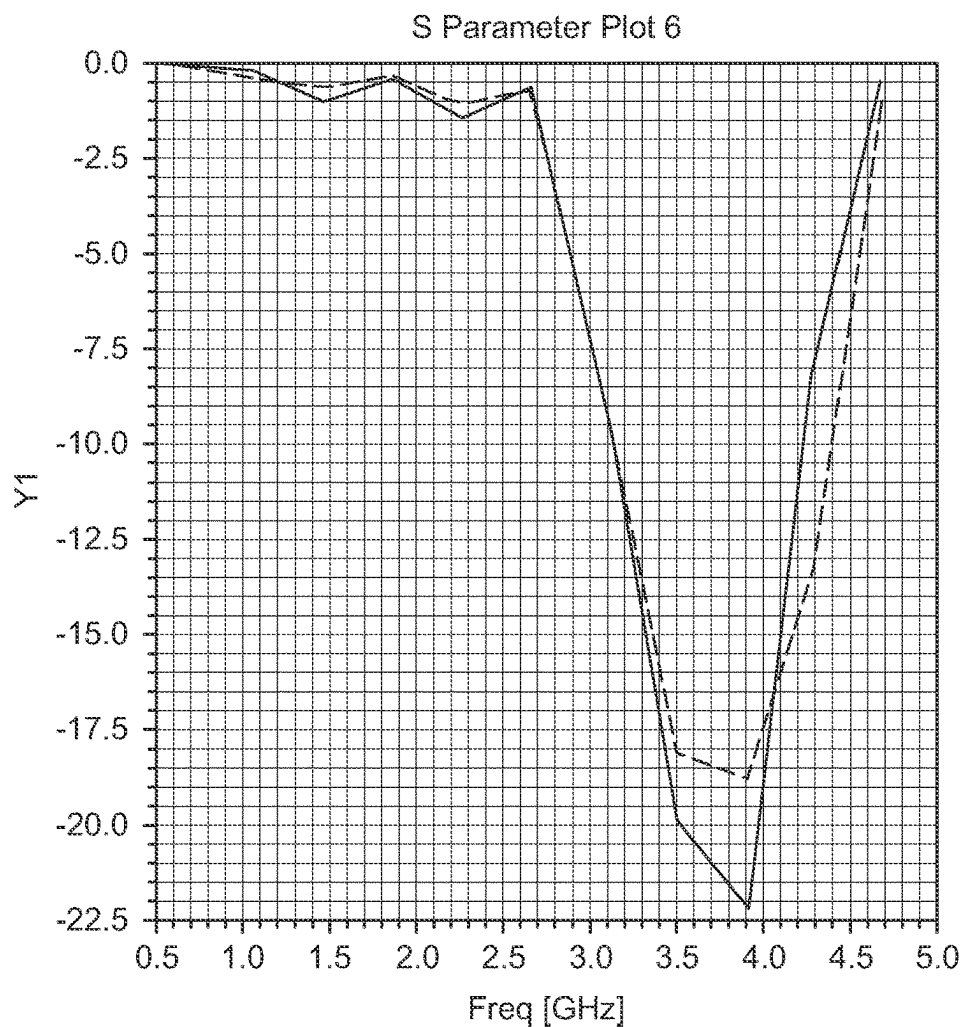
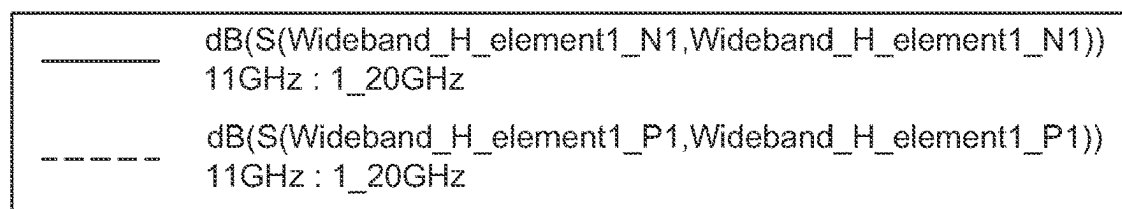
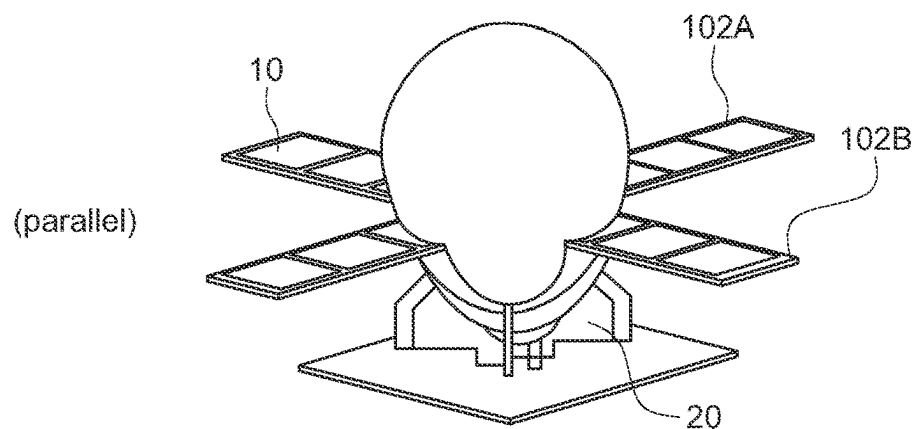


FIG. 16C

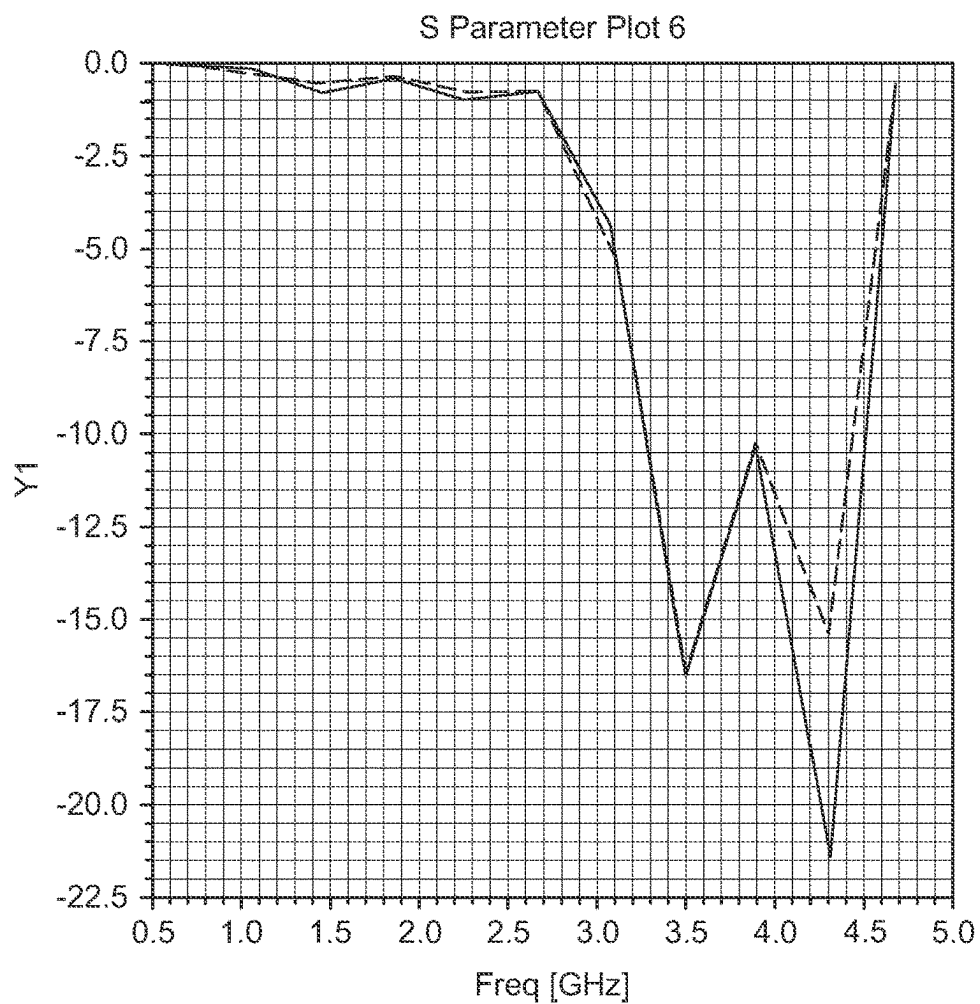
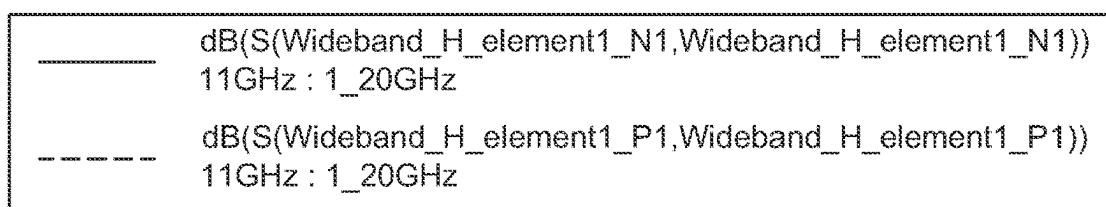
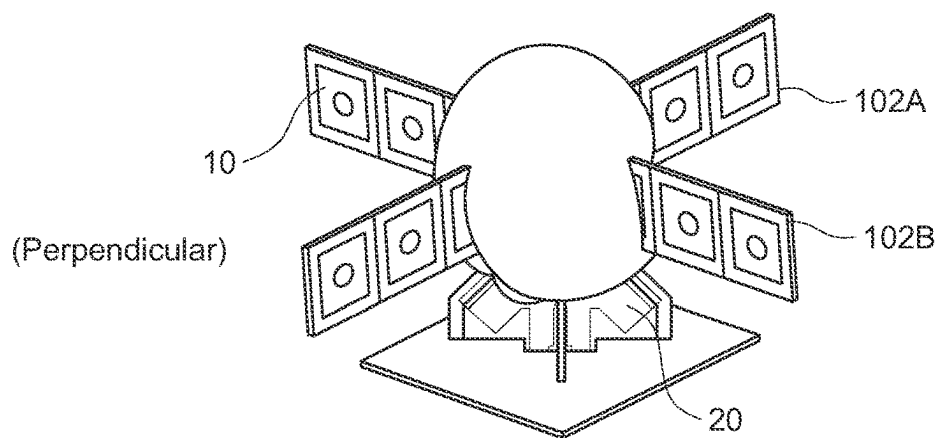


FIG. 16D

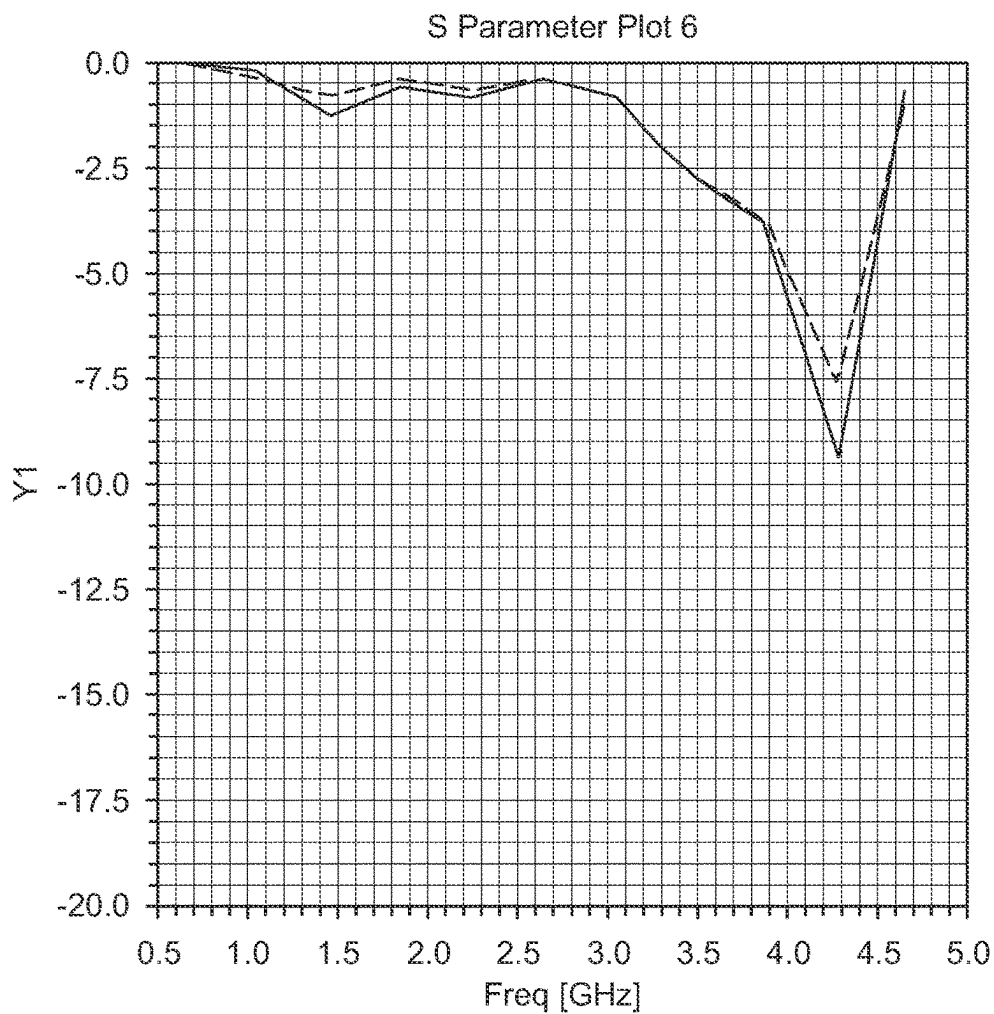
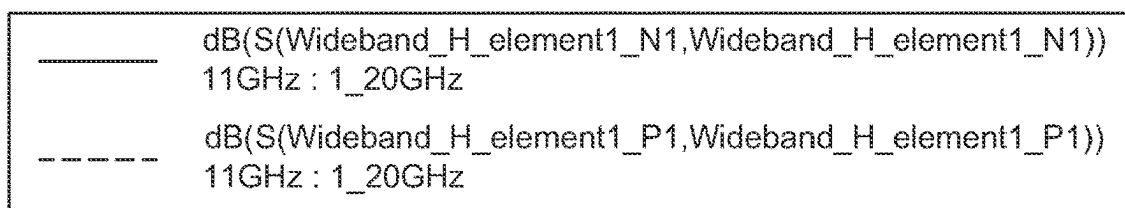
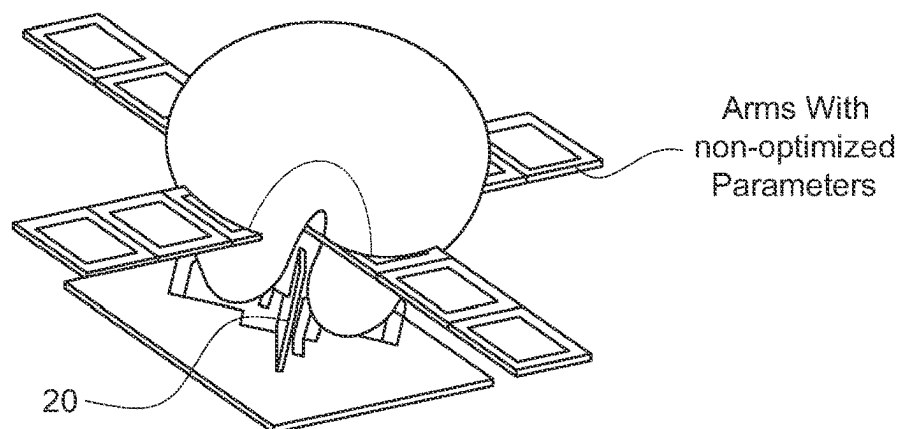


FIG. 16E

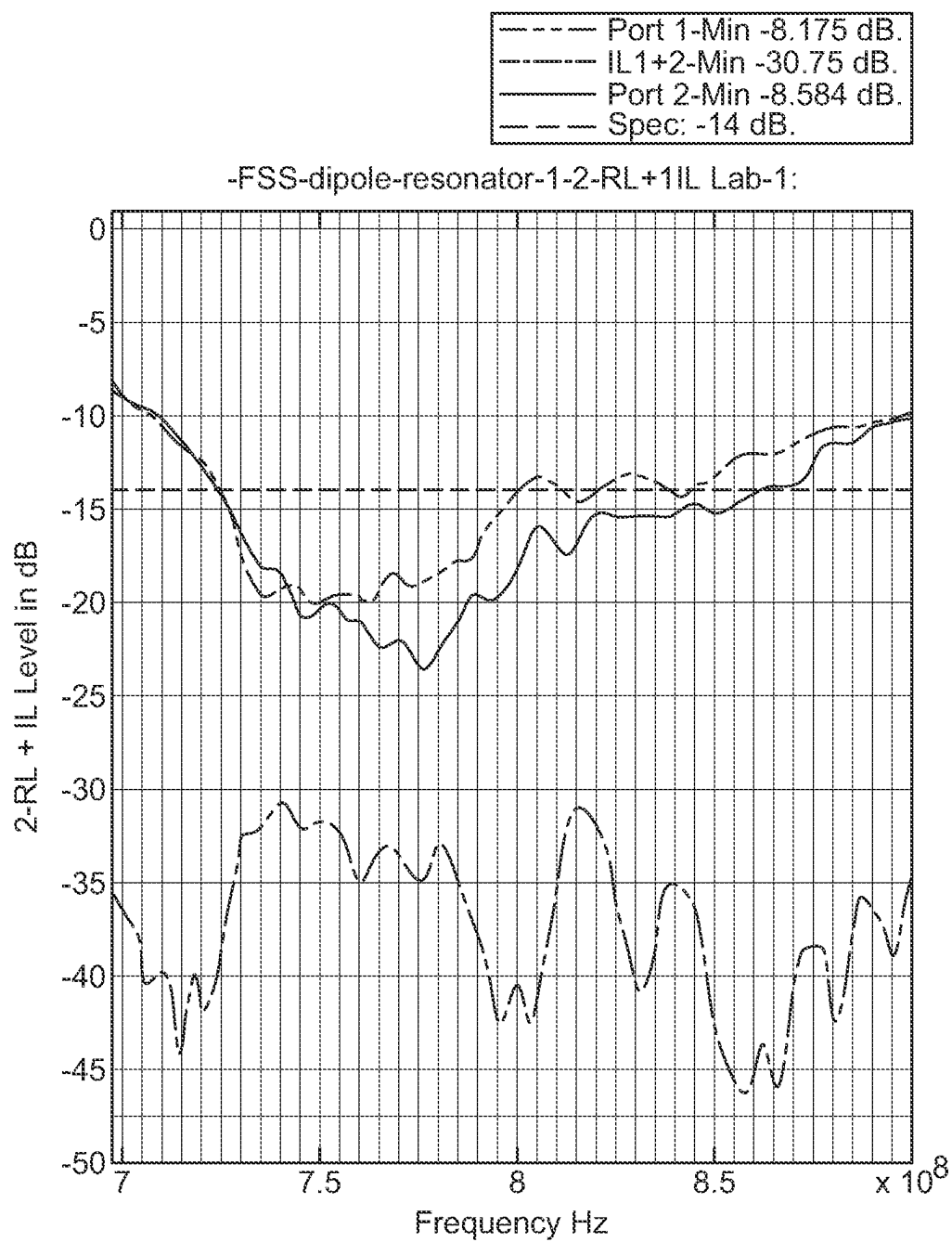


FIG. 17

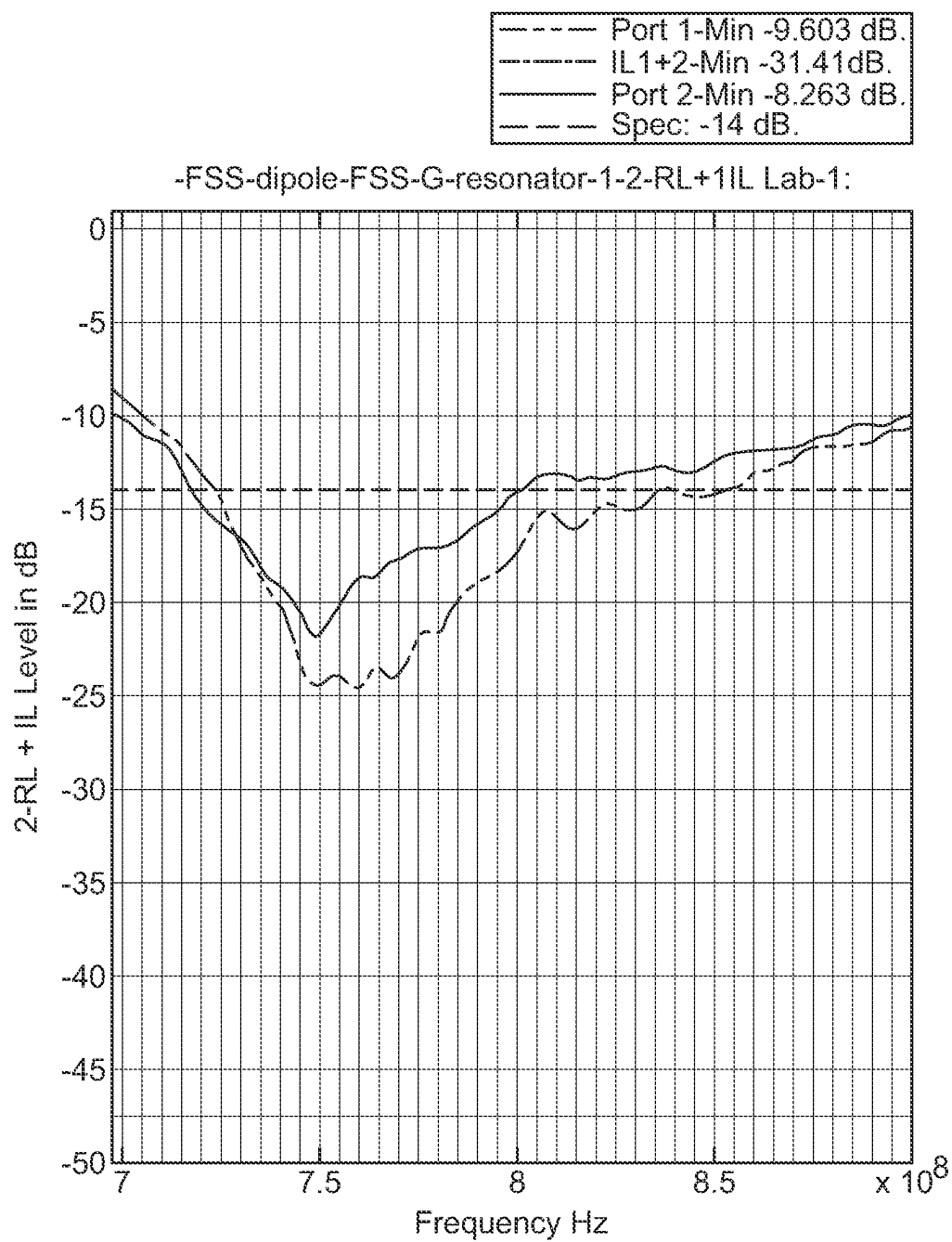


FIG. 18

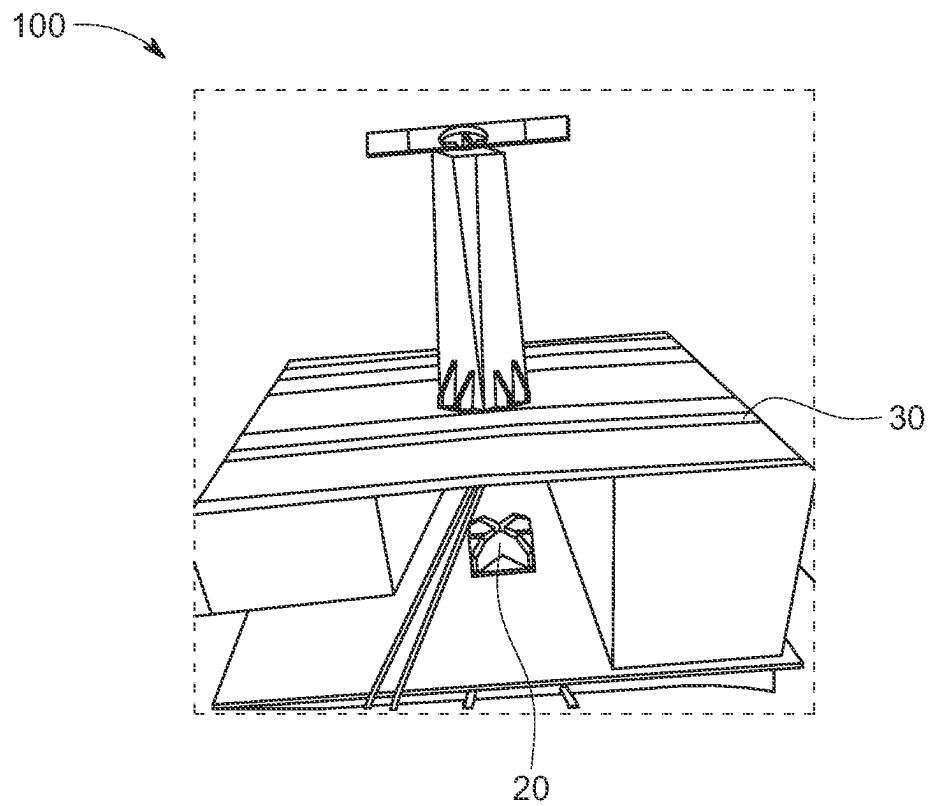


FIG. 19

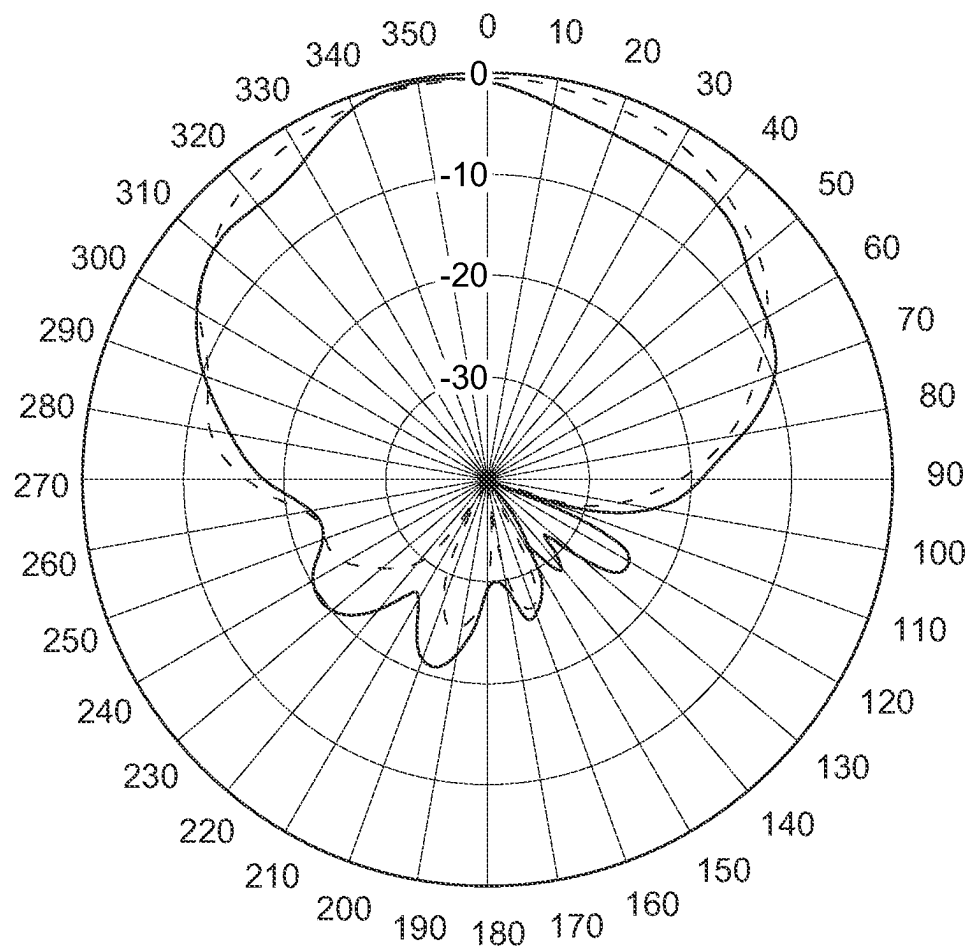


FIG. 20A

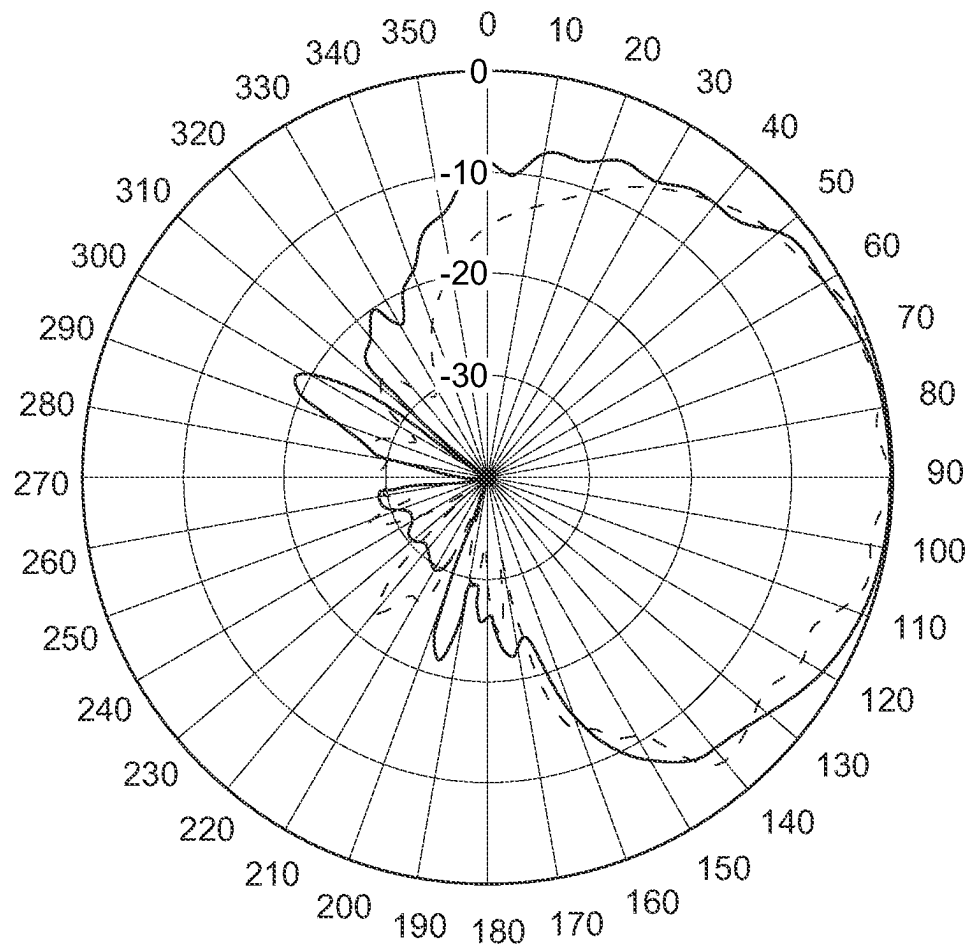


FIG. 20B

1

ANTENNA ELEMENT WITH SPATIAL FILTERING PROPERTY USING FREQUENCY SELECTIVE UNIT CELL BUILDING BLOCKS

RELATED APPLICATION

This application claims the benefit of priority from U.S. Provisional Patent Application No. 63/097,440 filed on Jan. 20, 2023, the entirety of which is incorporated by reference.

FILED OF THE INVENTION

This application relates to cellular base station antennas. More particularly, the present application relates to a novel construction for cellular base station antenna elements using Frequency Selective Surfaces (FSS).

DESCRIPTION OF RELATED ART

In the area of cellular base station antennas there are often low band (LB) elements (698-960 MHz) positioned on the same reflector as high band (HB) elements (e.g. 3.5 GHz) for example in a multi-band base station antenna. There are also situations that these LB and HB elements are located near each other but on two different reflector planes separated vertically. However, low band elements, typically in the form of dipoles (or any other elements such as patch elements), are larger than the high band elements and the physical structure of those LB elements interferes with the radiation pattern of the smaller high band elements positioned physically below them either on the same reflector or on another reflector underneath of the LB reflector.

In the prior art, when a plurality of LB and HB dipoles are located on a common reflector, different approaches were introduced to reduce coupling between the HB and LB dipoles. See for example U.S. Patent Publication No. US2021/0021037A1 each dipole arm has a plurality of conductive segments coupled in series by a plurality of inductive elements which attenuate or reduce induced current from high band elements (See Prior Art FIG. 1). In U.S. Pat. No. 9,698,486 matching circuits on the arms of higher band dipoles and feedboards are introduced to reduce their effect on lower band frequencies (See Prior Art FIG. 2 (showing FIG. 3 of the '486 patent)).

In another prior art arrangement shown in U.S. Pat. No. 11,387,567 dipole arms or feed lines of low band elements may include resonant structures to function as open circuit (s) for higher frequency bands but act as short circuit(s) for low band frequencies (See Prior Art FIG. 3).

All of these prior art designs use an approach for reduction or attenuation of coupling from first frequency band element such as a high band frequency element on the second frequency band element such as a low band element by introducing multiple resonators or a series of inductances.

OBJECTS AND SUMMARY

The present arrangement provides a novel approach to reduce coupling between BAND-1 and BAND-2 in a different way than the prior art. It is noted that BAND-1 in this application typically refers to a low frequency band (or LB) in the range of 698-890 MHz and BAND-2 in this application typically refers to a high band (or HB) in the range of 3300 MHz-3800 MHz. This is done for the sake of convenience,

2

but it is understood that the concepts referred to in this application are not limited by these particular frequency ranges.

In the present arrangement a novel construction is proposed for the LB (BAND-1) elements to reduce the amount of coupling with the HB (BAND-2) signal emanating from the high band element positioned nearby. In this context these two different band elements can be located on the same reflector or BAND-2 elements can be located on a different reflector underneath of the BAND-1 reflector.

In another embodiment, the BAND-2 elements located on a lower reflector are separated by a Frequency Selective Surface (FSS) from the BAND-1 dipole elements. The frequency selective surfaces (FSS), described in more detail below, are designed on a PCB with square patches with holes on one side and grids on the other side. This FSS surface acts as a reflector for LB BAND-1 dipole elements. In addition, the LB BAND-1 dipole structural components can be made either fully or partially from the same FSS materials as its reflector plate. This construction for the LB dipole structural elements reduces the effect of the BAND-1 low band dipole elements on the HB BAND-2 signal from the high band elements as the FSS unit structures used are almost "invisible" or "transparent" to the HB signal frequencies.

In another embodiment, the feed lines of the LB dipole, instead of traditional PCB circuit and balun feeds can be made from four (4) coaxial cables that are connected to the arms of the dipole on one side, with their external conductors being grounded to the FSS reflector on one of the square unit cells of the FSS. This compact feed structure further reduces the effect of the BAND-1 low band dipole elements on the HB BAND-2 signals emanating from the high band dipole elements.

As such, the present arrangement aims to provide a low band dipole unit that is constructed using an FSS "unit cell" concept, described in more detail below, that provides transparency for high-band dipole which improves pattern of high-band elements.

In the present arrangements an LB BAND-1 dipole is made using unit cells of a Frequency Selective surface (FSS). The unit cell is designed for passing 3300 MHz-3800 MHz (High Band: HB) and reflecting 698-890 MHz (Low-Band: LB). When this unit cell is used as arms of the LB dipole it reduces the HB signal currents on the arms of LB dipole considerably as the arms are almost invisible to the HB frequencies. In other words, the arms of a low-band dipole (BAND-1) act similar to metal for its own lower frequencies but is almost invisible to the high band radiation from the high band (BAND-2) elements positioned thereunder.

The FSS unit cells can be implemented in two different embodiments of a LB dipole. Thus, in one embodiment the present arrangement employs FSS unit cells are used as arms of the dipole as well as a simplified feed network for a dipole to reduce the effect of the feed network of the LB structure on HB signal pattern. The FSS arms for the LB dipole in this design may be arranged parallel to ground plane.

In another embodiment, FSS unit cells are used as arms of the dipole in place of the prior art metal arms in a traditional LB dipole element, but with a traditional PCB/balun feed network. The FSS arms for the LB dipole in this arrangement may be perpendicular to ground plane.

As noted above, it is noted that the arrangements described in terms of LB dipole elements, but the invention is not limited in this respect and can be used with other

frequency bands and rotation angles of arms with respect to ground plane or FSS plane, as explained below in the detailed description.

Additionally, the arrangement of the FSS unit cell arms of the LB dipole, being either parallel or perpendicular to the ground plane, may be both used in either the arrangement of the simplified feed network or may be used with the traditional balun/PCB feed network.

To this end a dipole element for use in a cellular base station antenna is provided. The dipole element has a feed network and at least two radiating arms at the top of said the network and arranged at an approximately 90 degree position relative to one another. The at least two radiating arms are constructed from a plurality of substantially identically dimensioned unit cells.

BRIEF DESCRIPTION OF THE DRAWINGS

The present invention can be best understood through the following description and accompanying drawing, wherein:

FIG. 1 is an image of a prior art low band dipole element; FIG. 2 is another image of a prior art of high-band radiating subarray;

FIG. 3 is another image of a prior art low band dipole element;

FIG. 4 is an image of a prior art antenna with two arrays and an FSS reflector positioned between the two;

FIG. 5A1 illustrates a top view of a unit cell in accordance with one embodiment;

FIG. 5A2 illustrates a bottom view of the unit cell of FIG. 5A1 in accordance with one embodiment;

FIG. 5A3 illustrates a top perspective view of the unit cell of FIG. 5A1 in accordance with one embodiment;

FIG. 5A4 illustrates a complete perspective view of the unit cell of FIG. 5A1 in accordance with one embodiment;

FIG. 5B shows the unit cell of FIGS. 5A1-5A4 in space, with the relevant dimensions and directional axes;

FIG. 5C is a return loss graph for a unit cell of FIG. 5A in accordance with one embodiment;

FIG. 6A is a FSS structure and HB dipole, in accordance with one embodiment;

FIG. 6B is an azimuth pattern of the HB dipole of FIG. 6A in accordance with one embodiment;

FIG. 7A shows the FSS structure and HB dipole from FIG. 6A with a prior art LB metal dipole positioned above;

FIG. 7B is an azimuth pattern of the HB dipole of FIG. 7A;

FIG. 8A shows a LB dipole over an FSS reflector in accordance with one embodiment;

FIG. 8B is a top view of the LB dipole of FIG. 8A in accordance with one embodiment;

FIG. 8C shows a feedboard baseplate of the LB dipole of FIG. 8A in accordance with one embodiment;

FIG. 8D shows the arms, in independent form, of the LB dipole of FIG. 8A in accordance with one embodiment;

FIG. 8E shows the arms of the LB dipole of FIG. 8A with current flow arrows in accordance with one embodiment;

FIG. 9A shows the LB dipole of FIG. 8A over a solid ground plane, in accordance with one embodiment;

FIG. 9B, shows a perspective view of the LB dipole and solid reflector of FIG. 9A in accordance with one embodiment;

FIG. 10A illustrates the LB dipole and feed network with FSS arms over an FSS plate above an HB dipole in accordance with one embodiment;

FIG. 10B is an azimuth pattern of the HB dipole of FIG. 10A in accordance with one embodiment;

FIG. 10C shows an LB dipole with the present feed cables, but prior art metal arms, over an FSS ground plane over an HB dipole;

FIG. 10D is an azimuth pattern of the HB dipole of FIG. 10C;

FIG. 11 is a graph of return loss of an LB dipole over a metal reflector and over an FSS reflector in accordance with one embodiment;

FIG. 12A illustrates an LB dipole with additional resonator structure, both made of unit cells, in accordance with one embodiment;

FIG. 12B is a return loss graph of the FSS dipole of FIG. 12A with no additional resonator, in accordance with one embodiment;

FIG. 12C is a return loss graph of the FSS dipole of FIG. 12A, in accordance with one embodiment;

FIG. 13 shows the FSS dipole with unit cell arms arranged perpendicular to the ground plane in accordance with one embodiment;

FIG. 14A shows an HB dipole under metal arm LB dipole (prior art) on a metal reflector;

FIG. 14B shows an HB dipole under a FSS LB dipole on a metal reflector in accordance with one embodiment;

FIG. 15 shows an HB azimuth pattern comparison of an HB dipole alone, as well as the HB arrangements in FIGS. 14A and 14B, in accordance with one embodiment; and

FIGS. 16A-16E shows pattern images and a return loss for five different cases regarding the signal pattern from a HB dipole element, in accordance with one embodiment;

FIG. 17 is a test graph of return loss of an LB dipole over a metal reflector in accordance with one embodiment;

FIG. 18 is a test graph of return loss of an LB dipole over a metal reflector and over an FSS reflector in accordance with one embodiment;

FIG. 19 shows illustrates a sample LB dipole and feed network with FSS arms over an FSS plate above an HB dipole in accordance with one embodiment;

FIG. 20A is an test azimuth pattern of the HB dipole of FIG. 19 in accordance with one embodiment; and

FIG. 20B is an test azimuth pattern of the HB dipole of FIG. 19 in accordance with one embodiment.

DETAILED DESCRIPTION

Beginning with the concept of FSS and “unit cells”, FSSs (Frequency Selective Surfaces) are periodic structures, mostly planar conductor arrays deposited on one side or both sides of a substrate which can be FR-4 (flame retardant woven glass-reinforced epoxy resin—PCB), RO4534 (R04000@ hydrocarbon ceramic laminates from the Rogers company) or similar substrates. See for example prior art U.S. Pat. No. 5,208,603, incorporated herein by reference. These conductor arrays are made of conductive shapes or apertures that can reflect or transmit electromagnetic radiation. These surfaces are used as spatial filters in electromagnetic applications and can pass specific frequency bands while reflecting other frequency bands. FSSs have periodic repetitions of a “particular structure” in two dimensions. This particular structure is called a “unit cell.” An example of this structure, implemented in the present arrangement, is a square ring on one side of substrate and a square patch with a hole on the other side of the substrate. Application of FSS principles for decoupling and for reusing the same aperture is known in base station antenna applications. Reusing antenna aperture means same physical space is used for two arrays by using FSS reflector in between. See for example prior art FIG. 4 taken from the IEEE 2019 publication

entitled “Decoupling and Low-Profile Design of Dual-Band Dual-Polarized Base Station Antennas Using Frequency-Selective Surface,” incorporated herein by reference.

For example the “Decoupling (IEEE 2019)” article discusses a dual-band dual-polarized base station antenna array that has shared-aperture between the two bands by introducing a frequency-selective surface (FSS) between radiators operating in the 0.69-0.96 GHz (B1) and 3.5-4.9 GHz (B2). In order to eliminate the influences of the mutual coupling on B1- and B2-band antennas, an FSS layer is introduced between B1 and B2 antenna elements. As can be seen from FIG. 4, the B2-band antenna elements are placed over the top layer. The B1-band patch is placed in the middle between an FSS layer and a ground plane. FSS acts as reflector for B2. Reflector for B1 is the ground plane shown in the figure. In prior art “Decoupling (IEEE 2019)” B2 is high-band and B1 is low-band. That means low-band elements are located under high-band elements which is in reverse to the present arrangement outlined below.

As noted above, these prior art unit cells are designed in a way that allow certain frequency bands to pass and stop other bands when they are implemented as a periodic structure. In particular in “Decoupling (IEEE 2019)” FSS passes the LB and acts as reflector for HB.

In the present arrangement, discussed below, using this basic concept of unit cells, a LB dipole is introduced that is based on using the unit cells of an FSS for its structural elements. In other words, instead of using an FSS layer as in prior art, the physical components of an LB dipole will be constructed using an FSS/unit cell structure. Although these unit cells are designed to work in a periodic structure, due to their capacitive/inductive characteristics, even by themselves, they can be used to generate a LB dipole structure that has a substantially lesser negative effect on the HB signal pattern compared to normal dipoles with metal wings. In other words, compared to prior art dipoles with metal arms, the present design for the LB dipole with unit cell arms is more transparent to an HB signal pattern from an adjacent HB element on the same reflector or on a reflector behind this reflector and separated by FSS layer. This can help in improving the aperture usage by placing higher band array below a an FSS low band dipole or below an FSS substrate with an FSS low band dipole above it as both the reflector and the LB dipole elements are transparent to HB frequencies.

Two exemplary implementation of such LB dipoles are discussed below:

- 1) An FSS based dual polarized LB dipole which has a light feed network with no PCB and balun and only 4 cables. The outer conductor of the cables are grounded to one of the unit cells of FSS surface. This dipole is designed with arms parallel to the ground plane; and
- 2) An LB dipole where the metal arms are replaced with FSS arms which are perpendicular to ground plane.

Starting with an explanation of the problem, in first step, an FSS unit cell **10** is shown, designed by simulation with following dimensions and following performance (FIGS. **5A1**, **5A2**, **5A3**, **5A4**, **5B** and **5C**). As can be seen in the return loss graph FIG. **5(C)** the design has a band pass for HB frequencies (e.g. -15.0 or less in the HB frequency range m1-m2).

Regarding the structure of unit cell **10**, FIG. **5A1** shows a top layer **11** of unit cell **10** constructed from copper in the form of a square ring. FIG. **5A2**, shows a bottom layer **12** of unit cell **10** constructed of copper in the form of solid square, with a central opening **13**. FIG. **5A3** shows a perspective view of a substrate layer **14** (with top top layer **11** thereon)

and having a central opening **15** that substantially matches the dimensions of central opening **13** in bottom layer **12**. FIG. **5A4** shows unit **10** with top layer **11** above substrate **14** as well as bottom layer **12** below substrate (it being understood that bottom layer **12** is shown in FIG. **5A4** for relative position but would not infact be visible from the top.

FIG. **5B** shows unit cell **10** in space with the relevant dimensions and directional axes. The exemplary values for the size of unit cell **10** in FIGS. **5A1**, **5A2**, **5A3**, **5A4** and **5B** are: $L=20.2$ mm, $d=5$ mm, $L_p=14.05$ mm, $w=0.625$ mm (see FIG. **5A** for L , d , L_p , and w). Unit cells **10** are thus squares with outer sides each equal to 20.2 mm. In this example the thickness of substrate **14** for unit cell **10** is 0.812 mm and is FR-4 material with dielectric constant of 4.4 . The dimensions of upper layer **11** in the form of a “square ring” has a width of 0.625 mm and on other side of substrate there is a square patch/bottom later **12**. This design for unit cell **10** is based on certain known unit cell/FSS applications, but is optimized for the present arrangement. For example, bottom layer **12** is a square where each side is 14.05 mm, which central opening **13** being a circle of 5 mm diameter.

With this arrangement a simulation was conducted testing the periodic surface made of unit cell **10** indicating that it passes HB frequencies and reflects LB frequencies. Thus, unit cell **10** is a new design over the prior art, relative to for example other types of grid patch FSS available in literature such as “Decoupling (IEEE 2019).”

For example, FIG. **6A** shows an HB dipole **20** under the designed FSS surface **30** using unit cells **10** such as shown in FIG. **5A**, and FIG. **6B** shows the azimuth pattern of HB dipole **20**. FSS surface **30** shown in this figure is based on unit cells **10** explained above. As can be seen from the signal pattern shape FSS surface **30** is transparent for HB signals emanating from HB dipole **20** thereunder. To this end, FSS surface **30** structure made with unit cells **10** is correctly sized and dimensioned so as not to impact HB patterns.

However, as noted above, the problem is that in some antenna designs there are LB dipoles located above HB elements **20** (ie above FSS surface **30** that HB element **20** is positioned under and the LB dipole is above). Therefore even though FSS structure **30** may be transparent for HB dipole radiation, the LB dipoles above FSS surface **30** are not transparent to HB dipole elements **20** and therefore the LB structure disturbs the radiation pattern of HB element **20**. For example, FIG. **7A** (prior art) shows H-band dipole **20** under an FSS surface with one prior art metal arm LB dipole element above FSS surface **30**, and FIG. **7B** (prior art) shows the azimuth pattern for HB frequencies: 3.3 GHz to 3.8 GHz. As can be seen by having a metal arm prior art LB dipole with a PCB feed board above FSS surface **30** and near an HB dipole **20** causes the pattern from HB dipole **20** to be significantly disturbed.

The present arrangement provides a novel LB dipole **100** with arms designed and implemented from the same type of FSS/unit cell **10** construction which is used for the LB dipoles FSS ground plane **30** as shown in FIGS. **8A-8D**. LB dipole **100** has two arms **102A** and **102B** (shown independently in FIG. **8D**, a base plate **104** (shown independently in FIG. **8C**), with arms **102A** and **102B** connected in a normal cross shape for dipole arms. As shown in FIGS. **8A-8D**, arms **102A** and **102B** and feedboard base plate **104** are constructed from unit cells **10**.

As shown in FIGS. **8A** and **8B** as well as FIG. **9B** below, the feedlines of LB dipole **100** are in the form of cables **106** with direct connection to dipole arms **102A** and **102B** that additionally provide less signal blockage of HB frequencies compared to traditional PCB feedlines and located on one of

unit cells **10** as shown for Example in FIGS. **8C** and **8D**. This design for feed cables **106** can be implemented using both 75 ohm and 50 ohm cables with an exemplary 3 mm outside diameter, however it is not limited to these impedances.

As shown in FIG. **8D** arms **102A** and **102B** each illustrate six (6) unit cells **10** each, three (3) on either side of a central connection piece **112**. In FIG. **8D**, dipole arm **102A** is shown in “bottom” view meaning that bottom layer **12** is the upward facing side of arm **102A**. However, dipole arm **102b** is shown in “top” view meaning that top layer **11** is the upward facing side of arm **102B**. As such, arms **102** may be assembled and configured into LB dipole **100** with each of unit cells **10** having upper layer **11** facing upward (**102B**) away from reflector **30** or may be assembled and configured into LB dipole **100** with each of unit cells **10** having lower layer **12** facing upward (**102A**) away from reflector **30**. The arrangement for both arms **102** of dipole **100** can be either both lower layer **12** facing upward (**102A**), both upper layer **11** facing upward (**102B**) or one of each depending on the desired top or bottom polarization although there may typically no noticeable difference in performance between the three options.

As show in FIG. **8E**, illustrating one exemplary arm **102B** (upper layer **11** upwardly facing), the signal current coming from feed lines **106** connects at central connector **112** and passes along the continuous metal grid formed by upper layer **11** of each of unit cells **10**.

In one embodiment FIGS. **9A** and **9B** show two different perspectives of the present FSS LB dipole **100** design and its feed cables **106**. Unlike the prior art, there is no PCB or balun, but instead four cables **106** directly go through one unit cell **10** (e.g. through openings **108** shown in FIG. **8C**) and connecting with a soldermask openings **110** on central connection piece **112** on arms **102A/102B** (e.g. as shown in FIG. **8D**).

FIG. **10A** shows LB dipole **100** positioned over FSS layer **30** with HB element **20** positioned under FSS layer **30** and FIG. **10B** shows the HB azimuth signal pattern of HB element **20**. As can be seen the pattern distortion is much less than prior art antennas of similar design (e.g. prior art FIG. **7B**) and is more similar to the original pattern (FIG. **6B** in the absence of any LB dipole). To confirm that better signal pattern in FIG. **10B** is at least partly due to the using of FSS/unit cell constructed arms in LB dipole **100**, as opposed to being attributable only to feed cables **106**, another simulation is run with LB dipole arms **102A/102B** replaced by prior art metal as shown in FIG. **10C** (but with the new cable feeds **106**), and as can be observed in FIG. **10D**, the HB azimuth pattern is degraded relative to the pattern in FIG. **10B** relating to the present arrangement.

It should be mentioned that the new FSS LB dipole **100** radiation and RL (return loss) stays the same above either the present FSS reflector surface **30** and/or a basic metal reflector surface, which shows that FSS surface **30** acts similar to metal with respect to the radiation from LB dipole element **100**. This demonstrates that LB dipole **100**, even with the present novel constructions, is working well in its own right as an LB element, in addition to being transparent to the HB signal pattern. For example, FIG. **11** shows the RL of FSS LB dipole **100** of FIG. **8A** over a metal reflector as well as over FSS layer **30** for one implementation (e.g. 75 ohm feed).

In some implementations, to increase or adjust the bandwidth of the present LB FSS dipole **100** an extra resonator **120** can be added, e.g. as rings, cross or patches. As with LB dipole **100**, resonator **120** can likewise be made of FSS unit cells **10**. For example, FIG. **12A** shows an image of LB

dipole **100** with an added resonator **120** located 10 mm above the main resonator arms **102A** and **102B** to improve the RL (return loss) in the higher part of the low band frequency range. As shown in FIG. **12A** both arms of added resonator **120** are shown in this exemplary arrangement with unit cells arranged with upper layer **11** upwardly facing. In the present example, resonator **120** can be made of five-unit cells **10**. FIG. **12B** shows the RL of both polarizations of LB FSS dipole **100** with no resonator **120** and FIG. **12C** shows the RL of both polarizations of LB dipole **100** with added resonator **120**.

In another embodiment, as a simple way to make FSS LB dipole **100**, metal arms of a standard LB dipole are replaced by FSS cell unit(s) **10** as explained above. FIG. **13** shows dipole **100** with vertical FSS arms **102A/102B**. As shown in FIG. **13**, here the embodiment includes two arms **102A** and **102B** turned vertically (i.e. perpendicular to ground plane). Also this version of LB dipole **100** does not have cable feeds **106** as in FIGS. **8** and **9**, but instead uses a normal PCB balun **125**. Additionally, this version of LB dipole **100** maintains added resonator **120**.

A test was run (three runs) the first being with HB dipole **20** positioned on a normal metal reflector by itself as reference and then the other two tests were run with HB dipole **20** on a metal reflector under a prior art LB dipole (FIG. **14A**) and then again with HB dipole **20** on the metal reflector under FSS LB dipole **100** (FIG. **14B**).

As can be seen from FIG. **15**, with dipole **100** with FSS arms **102A/102B**, the azimuth pattern from HB element **20** is more similar to the case with no lowband dipole at all than it is when positioned under a prior art metal armed dipole, the later demonstrating a sharp dip in the HB pattern. This means that arms **102A/102B** being made of unit cells **10** has less an effect on the HB signal patterns from a underlying HB element **20** than prior art metal arms. This is true even with the orientation of the FSS arms **102A** and **102B** being vertical or perpendicular to the reflector as shown in FIG. **14B** (and FIG. **13**). This supports that arms **102A** and **102B** made of unit cells **10** is effective in both parallel (FIG. **10B**) and perpendicular (FIG. **15**) orientations.

To further show the transparency of FSS LB dipole **100** to the HB pattern from HB element **20** positioned thereunder five different cases are compared.

In a first case there is a single HB dipole **20** radiating. In the second case there is a metal cross (which has a similar shape to an LB dipole) at a very short distance (8 mm) above an HB element. The third case considers FSS cross **102A/102B** positioned horizontally above HB dipole **20** (using optimized parameters: $L=20.2$ mm, $d=5$ mm, $L_p=14.05$ mm, and $w=0.625$ mm). The fourth case considers two FSS arms **102A/102B** (with designed parameters) positioned vertically above HB dipole **20**. The fifth case considers an FSS cross **102A/102B** above HB dipole **20** using random parameters: $L=20.2$ mm, $d=2$ mm, $L_p=9.2$ mm, $w=3$ mm i.e. with “unit cells” that are not optimized to pass HB frequencies and radiate LB frequencies. It is noted that the FSS unit cell **10** design parameters (L , d , L_p , and W) are preferably optimized for being transparent to HB and act similar to metal for LB. However, as shown in FIG. **16E**, if the design parameters of unit cells **10** are selected arbitrarily even with the same shape the FSS arms **102A/102B** it will be less effective.

As can be seen from FIGS. **16A-16E**, pattern and RL (return loss) of HB dipole **20** is completely deteriorated in second case (B) (ie similar to prior art metal dipole) compared to first case (A). While both the third (C) and fourth (D) cases show minimum change compared to pattern of HB

9

dipole **20** itself and RL is >10 dB in the required HB (3300-3800 MHz) showing that the present FSS dipole arms **102A/102B** do not significantly effect the HB pattern or RL in both parallel and perpendicular orientations. The fifth case (E) demonstrates that when the design parameters of the unit cell/FSS **10** are selected arbitrarily then the HB pattern, and particularly the return loss, are degraded.

It is noted that above measurements and examples were tested in a proof of concept experiment. For example FIG. **11** shows the RL of FSS LB dipole **100** of FIG. **8A** over a metal reflector as well as over FSS layer **30** for one implementation (e.g. 75 ohm feed). FIG. **17** shows an actual measurement of return loss FSS LB dipole **100** of FIG. **8A** over a metal reflector. As shown in FIG. **17**, the return loss is greater than 10 dB almost over the entire band, similar to the line in FIG. **11** for the metal reflector line. Similarly, FIG. **18** shows an actual measurement of return loss FSS LB dipole **100** of FIG. **8A** but over an FSS layer **30**. As shown in FIG. **18**, the return loss is greater than 10 dB almost over the entire band, similar to the line in FIG. **11** for the FSS reflector line.

FIG. **19** is an exemplary construction of FSS dipole **100** positioned on FSS layer **30** over HB dipole **20**, similar to the schematic shown in FIG. **10A**. As expected, FIG. **20A** shows the azimuth signal pattern of HB element **20** similar to that in the model pattern shown in FIG. **10B** with much less distortion than prior art antennas of similar design (e.g. prior art FIG. **7B**) and more similar to the original pattern (FIG. **6B** in the absence of any LB dipole). FIG. **20b** shows the elevation pattern of HB dipole **20** from FIG. **19** again with low distortion, showing that it too is not significantly effected by FSS LB dipole **100**.

While only certain features of the invention have been illustrated and described herein, many modifications, substitutions, changes or equivalents will now occur to those skilled in the art. It is therefore, to be understood that this application is intended to cover all such modifications and changes that fall within the true spirit of the invention.

What is claimed is:

1. An antenna element for use in a cellular base station antenna, the antenna element comprising:

a feed network; and

at least two radiating arms at the top of said feed network and arranged at an approximately 90 degree position relative to one another,

wherein said at least two radiating arms are constructed from a plurality of identically dimensioned unit cells, each of said unit cells are made from a square dielectric substrate with a square metal ring on a perimeter of one side of the substrate, and a square metal patch on another side of the substrate,

wherein said square metal patch has a round hole in the middle, and

wherein said square dielectric substrate has a central opening in the middle that matches the dimensions of the round hole in said square metal patch.

10

2. The antenna element as claimed in claim 1, wherein said antenna element is configured to support a signal range of 698-890 MHz.

3. The antenna element as claimed in claim 2, wherein said unit cells of said radiating arms are constructed so to allow signals in the range of 3300 MHz-3800 MHz to pass therethrough.

4. The antenna element as claimed in claim 1, wherein said square dielectric substrate is about 0.812 mm thick and has a side length of 20.2 mm.

5. The antenna element as claimed in claim 4, wherein said square ring on the perimeter of one side of said substrate has width of 0.625 mm per side.

6. The antenna element as claimed in claim 4, wherein said square patch on said another side of said substrate is a square metal patch with a side length of 14.05 mm, with the round hole in the middle having a 5 mm diameter.

7. The antenna element as claimed in claim 1, wherein said antenna element is positioned above a ground plane and wherein said at least two radiating arms are arranged perpendicular to said ground plane.

8. The antenna element as claimed in claim 1, wherein said antenna element is positioned above a ground plane and wherein said at least two radiating arms are arranged parallel to said ground plane.

9. The antenna element as claimed in claim 1, wherein said antenna element further comprises a resonator above said at least two radiating arms.

10. The antenna element as claimed in claim 9, wherein said resonator is made of unit cells.

11. The antenna element as claimed in claim 3, wherein said antenna element is an antenna element radiating at 698-890 MHz included in an antenna arrangement on a metal reflector with at least one other antenna element on the metal reflector positioned thereunder that radiates in the range of 3300 MHz-3800 MHz.

12. The antenna element as claimed in claim 3, wherein said antenna element is an antenna element radiating at 698-890 MHz included in an antenna arrangement on a frequency selective surface reflector with at least one other element positioned under the frequency selective surface reflector, that radiates in the range of 3300 MHz-3800 MHz.

13. The antenna element as claimed in claim 12, wherein said frequency selective surface reflector is made from unit cells.

14. The antenna element as claimed in claim 1, wherein said antenna element further comprises a printed circuit board balun for feeding signals to said at least two radiating arms.

15. The antenna element as claimed in claim 1, wherein said antenna element further comprises a plurality of vertically arranged signal cables for feeding signals to said at least two radiating arms, said signal cables grounded to one of said unit cells of said frequency selective surface reflector.

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